20

Infinite Series

20.1 SEQUENCE

A *sequence* is a succession of numbers or terms formed according to some definite rule. The nth term in a sequence is denoted by u_n.

For example, if $u_n = 2n + 1$.

By giving different values of n in u_n, we get different terms of the sequence.

Thus, $u_1 = 3,\ u_2 = 5,\ u_3 = 7 \ldots$

A sequence having unlimited number of terms is known as an *infinite sequence*.

20.2 LIMIT

If a sequence tends to a limit l, then we write

$$\lim_{n \to \infty} (u_n) = l$$

20.3 CONVERGENT SEQUENCE

If the limit of a sequence is finite, the sequence is *convergent.* If the limit of a sequence does not tend to a finite number, the sequence is said to be *divergent*.

e.g., $1,\ \frac{1}{4},\ \frac{1}{9},\ \frac{1}{16},\ ...,\ \frac{1}{n^2} + \ldots$ is a convergent sequence.

$3,\ 5,\ 7,\ ...,\ (2n + 1),\ \ldots$ is a divergent sequence.

20.4 BOUNDED SEQUENCE

$u_1,\ u_2,\ u_3,\ ...,\ u_n\ \ldots$ is a bounded sequence if $u_n < k$ for every n.

20.5 MONOTONIC SEQUENCE

The sequence is either increasing or decreasing, such sequences are called *monotonic*.

e.g., 1, 4, 7, 10, ... is a monotonic sequence.

$1,\ \frac{1}{2},\ \frac{1}{3},\ \frac{1}{4},\ \ldots$ is also a monotonic sequence.

$1,\ -1,\ 1,\ -1,\ 1,\ \ldots$ is not a monotonic sequence.

A sequence which is monotonic and bounded is a convergent sequence.

Exercise 20.1

Determine the general term of each of the following sequence. Prove that the following sequences are convergent.

1. $\frac{1}{2},\ \frac{1}{4},\ \frac{1}{8},\ \frac{1}{16},\ \ldots$ **Ans.** $\frac{1}{2^n}$

2. $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \ldots$ **Ans.** $\frac{n}{n+1}$

3. $1, -1, 1, -1, \ldots$ **Ans.** $(-1)^{n-1}$

4. $\frac{1^2}{\lfloor 1}, \frac{2^2}{\lfloor 2}, \frac{3^2}{\lfloor 3}, \frac{4^2}{\lfloor 4}, \frac{5^2}{\lfloor 5}, \ldots$ **Ans.** $\frac{n^2}{\lfloor n}$

Which of the following sequences are convergent ?

5. $u_n = \frac{n+1}{n}$ **Ans.** Convergent

6. $u_n = 3n$ **Ans.** Divergent

7. $u_n = n^2$ **Ans.** Divergent

8. $u_n = \frac{1}{n}$ **Ans.** Convergent

20.6 REMEMBER THE FOLLOWING LIMITS

(*i*) $\lim_{n\to\infty} x^n = 0$ if $x < 1$ and $\lim_{n\to\infty} x^n = \infty$ if $x > 1$ (*ii*) $\lim_{n\to\infty} \frac{x^n}{n!} = 0$ for all values of x

(*iii*) $\lim_{n\to\infty} \frac{\log n}{n} = 0$ (*iv*) $\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e$ (*v*) $\lim_{n\to\infty} (n)^{1/n} = 1$

(*vi*) $\lim_{n\to\infty} [n!]^{1/n} = \infty$ (*vii*) $\lim_{n\to\infty} \left[\frac{(n!)}{n}\right]^{1/n} = \frac{1}{e}$ (*viii*) $\lim_{n\to\infty} n x^n = 0$ if $x < 1$

(*ix*) $\lim_{n\to\infty} n^h = \infty$ (*x*) $\lim_{n\to\infty} \frac{1}{n^h} = 0$

(*xi*) $\lim_{x\to 0}\left[\frac{a^x - 1}{x}\right] = \log a$ or $\lim_{n\to\infty} \frac{a^{1/n} - 1}{1/n} = \log a$ (*xii*) $\lim_{x\to 0} \frac{\sin x}{x} = 1$ (*xiii*) $\lim_{x\to 0} \frac{\tan x}{x} = 1$

20.7 SERIES

A *series* is the sum of a sequence.

Let $u_1, u_2, u_3, \ldots\ldots\ldots\ldots, u_n, \ldots\ldots\ldots\ldots$ be a given sequence. Then, the expression

$u_1 + u_2 + u_3 + \ldots\ldots\ldots\ldots + u_n + \ldots\ldots\ldots\ldots$ is called the series associated with the given sequence.

For example, $1 + 3 + 5 + 7 + \ldots$ is a series.

If the number of terms of a series is limited, the series is called *finite*. When the number of terms of a series are unlimited, it is called an *infinite series*.

$$u_1 + u_2 + u_3 + u_4 + \ldots + u_n + \ldots \infty$$

is called an infinite series and it is denoted by $\sum_{n=1}^{\infty} u_n$ or Σu_n. The sum of the first n terms of a series is denoted by S_n.

20.8 CONVERGENT, DIVERGENT AND OSCILLATORY SERIES

Consider the infinite series $\Sigma u_n = u_1 + u_2 + u_3 + \ldots + u_n + \ldots \infty$

$$S_n = u_1 + u_2 + u_3 + \ldots + u_n$$

Three cases arise :

(*i*) If S_n tends to a finite number as $n \to \infty$, the series Σu_n is said to be *convergent.*

(*ii*) If S_n tends to infinity as $n \to \infty$, the series Σu_n is said to be *divergent.*

(*iii*) If S_n does not tend to a unique limit, finite or infinite, the series Σu_n is called *oscillatory.*

20.9 PROPERTIES OF INFINITE SERIES

1. The nature of an infinite series does not change :

(*i*) by multiplication of all terms by a constant *k*.

(*ii*) by addition or deletion of a finite number of terms.

2. If two series Σu_n and Σv_n are convergent, then $\Sigma (u_n + v_n)$ is also convergent.

Example 1. *Examine the nature of the series* $1 + 2 + 3 + 4 + \ldots + n + \ldots \infty$.

Solution. $S_n = 1 + 2 + 3 + 4 + \ldots + n = \dfrac{n(n+1)}{2}$ [Series in A.P.]

since $\lim\limits_{n \to \infty} S_n = \lim\limits_{n \to \infty} \dfrac{n(n+1)}{2} \Rightarrow \infty$

Hence, this series is divergent.

Example 2. *Test the convergence of the series* $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots \infty$.

Solution. $S_n = 1 + \dfrac{1}{2} + \dfrac{1}{4} + \dfrac{1}{8} + \ldots \infty$ [Series in G.P.]

$= \dfrac{1}{1 - \frac{1}{2}} = 2 \qquad \left(S_n = \dfrac{a}{1-r} \right)$

$\lim\limits_{n \to \infty} S_n = 2$

Hence, the series is convergent.

Example 3. *Discuss the nature of the series* $2 - 2 + 2 - 2 + 2 - \ldots \infty$.

Solution. $S_n = 2 - 2 + 2 - 2 + 2 - \ldots \infty$

$= 0$ if n is even.

$= 2$ if n is odd.

Hence, S_n does not tend to a unique limit, and, therefore, the given series is oscillatory.

Exercise 20.2

Discuss the nature of the following series :

1. $1 + 4 + 7 + 10 + \ldots \infty$ **Ans.** Divergent

2. $1 + \frac{5}{4} + \frac{6}{4} + \frac{7}{4} + \ldots \infty$ **Ans.** Divergent

3. $6 - 5 - 1 + 6 - 5 - 1 + 6 - 5 - 1 + \ldots \infty$ **Ans.** Oscillatory

4. $3 + \dfrac{3}{2} + \dfrac{3}{2^2} + \ldots \infty$ **Ans.** Convergent

5. $1^2 + 2^2 + 3^2 + 4^2 + \ldots \infty$ **Ans.** Divergent

6. $1 - \dfrac{1}{2} + \dfrac{1}{4} - \dfrac{1}{8} + \ldots \infty$ **Ans.** Convergent

7. $\dfrac{1}{1 \cdot 3} + \dfrac{1}{3 \cdot 5} + \dfrac{1}{5 \cdot 7} + \ldots \infty$ **Ans.** Convergent

8. $\dfrac{1}{\lfloor 2} + \dfrac{2}{\lfloor 3} + \dfrac{3}{\lfloor 4} + \ldots \infty$ **Ans.** Convergent

9. $\log 3 + \log \dfrac{4}{3} + \log \dfrac{5}{4} + \ldots \infty$ **Ans.** Divergent

10. $\sum \log \dfrac{n}{n+1}$ **Ans.** Divergent

11. $\Sigma (\sqrt{n+1} - \sqrt{n})$ **Ans.** Divergent

12. $\sum \dfrac{1}{n(n+2)}$ **Ans.** Convergent

13. $\sum \dfrac{1}{n(n+1)(n+2)(n+3)}$ **Ans.** Convergent

14. $\sum \dfrac{n}{(n+1)(n+2)(n+3)}$ **Ans.** Convergent

15. $\sum \dfrac{2n+1}{n^2 (n+1)^2}$ **Ans.** Convergent

20.10 PROPERTIES OF GEOMETRIC SERIES

The series $1 + r + r^2 + r^3 + \ldots \infty$ is

(*i*) convergent if $|r| < 1$ (*ii*) divergent if $r \geq 1$ (*iii*) oscillatory if $r \leq -1$.

Proof. $S_n = 1 + r + r^2 + \ldots + r^{n-1} = \dfrac{1 - r^n}{1 - r}$

(*i*) When $|r| < 1$, $\lim\limits_{n \to \infty} r^n = 0$

$$\lim_{n \to \infty} S_n = \lim_{n \to \infty} \frac{1 - r^n}{1 - r} = \frac{1 - 0}{1 - r} = \frac{1}{1 - r}$$

Hence, the series is convergent.

(*ii*) (*a*) When $r > 1$, $\lim\limits_{n \to \infty} r^n = \infty$ $\quad \therefore \lim\limits_{n \to \infty} S_n = \lim\limits_{n \to \infty} \dfrac{r^n - 1}{r - 1} \Rightarrow \infty$

Hence, the series is divergent.

(*b*) When $r = 1$, the series becomes $1 + 1 + 1 + 1 + \ldots \infty$

$$S_n = 1 + 1 + 1 + 1 + \ldots = n$$

$$\lim_{n \to \infty} S_n = \lim_{n \to \infty} n = \infty$$

Hence, the series is divergent.

(*iii*) (*a*) When $r = -1$, the series becomes $1 - 1 + 1 - 1 + 1 - \ldots \infty$

$$S_n = 0 \text{ if } n \text{ is even}$$

$$= 1 \text{ if } n \text{ is odd}$$

Hence, the series is oscillatory.

(*b*) When $r < -1$, let $r = -k$ where $k > 1$.

$$r^n = (-k)^n = (-1)^n k^n$$

$$\lim_{n \to \infty} S_n = \lim_{n \to \infty} \frac{1 - r^n}{1 - r} = \lim_{n \to \infty} \frac{1 - (-1)^n k^n}{1 - (-k)}$$

$$= +\infty \text{ if } n \text{ is odd}$$

$$= -\infty \text{ if } n \text{ is even}$$

Hence, the series is oscillatory.

Exercise 20.3

Test the nature of the following series :

1. $1 + \dfrac{1}{2} + \dfrac{1}{2^2} + \dfrac{1}{2^3} + \ldots \infty$ **Ans.** Convergent

2. $1 + \dfrac{3}{4} + \dfrac{9}{16} + \dfrac{27}{64} + \ldots \infty$ **Ans.** Convergent

3. $1 - \dfrac{1}{3} + \dfrac{1}{9} - \dfrac{1}{27} + \ldots \infty$ **Ans.** Convergent

4. $1 - 2 + 4 - 8 + \ldots \infty$ **Ans.** Oscillatory

5. $2 + 3 + \dfrac{9}{2} + \dfrac{27}{4} + \ldots \infty$ **Ans.** Divergent

6. $1 + \dfrac{4}{3} + \left(\dfrac{4}{3}\right)^2 + \left(\dfrac{4}{3}\right)^3 + \ldots \infty$ **Ans.** Divergent

7. State, which one of the alternatives in the following, is correct :

The series $1 - 1 + 1 - 1 + \ldots$ is

(*i*) Convergent with its sum equal to 0. (*ii*) Convergent with its sum equal to 1.

(*iii*) Divergent. (*iv*) Oscillatory. **Ans.** Oscillatory series

20.11 POSITIVE TERM SERIES

If all terms after few negative terms in an infinite series are positive, such a series is a positive term series.

e.g., $-10-6-1+5+12+20+\ldots$ is a positive term series.

By omitting the negative terms, the nature of a positive term series remains unchanged.

20.12 NECESSARY CONDITIONS FOR CONVERGENT SERIES

For every convergent series Σu_n,

$$\lim_{n\to\infty} u_n = 0$$

Solution. Let $S_n = u_1 + u_2 + u_3 + \ldots + u_n$

$$\lim_{n\to\infty} S_n = k \qquad \text{(a finite quantity)}$$

Also $$\lim_{n\to\infty} S_{n-1} = k \qquad \text{(a finite quantity)}$$

$$S_n = S_{n-1} + u_n$$

$$u_n = S_n - S_{n-1}$$

$$\lim_{n\to\infty} u_n = \lim_{n\to\infty} [S_n - S_{n-1}] = 0$$

$$\lim_{n\to\infty} u_n = 0$$

Corollary. Converse of the above theorem is not true.

e.g., $1+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\frac{1}{\sqrt{4}}+\ldots+\frac{1}{\sqrt{n}}+\ldots\infty$ is divergent.

$$S_n = 1+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\frac{1}{\sqrt{4}}+\ldots+\frac{1}{\sqrt{n}} > \frac{1}{\sqrt{n}}+\frac{1}{\sqrt{n}}+\frac{1}{\sqrt{n}}+\frac{1}{\sqrt{n}}+\ldots+\frac{1}{\sqrt{n}}$$

$$> \frac{n}{\sqrt{n}} > \sqrt{n}$$

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \sqrt{n} = \infty$$

Thus, the series is divergent, although $\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$

So $\lim_{n\to\infty} u_n = 0$ is a necessary condition but not a sufficient condition for convergence.

Note : 1. Test for divergence

If $\lim_{n\to\infty} u_n \neq 0$, *the series* Σu_n *must be divergent.*

2. *To determine the nature of a series we have to find* S_n. *Since it is not possible to find* S_n *for every series, we have to devise tests for convergence without involving* S_n.

20.13 CAUCHY'S FUNDAMENTAL TEST FOR DIVERGENCE

If $\lim_{n\to\infty} u_n \neq 0$, the series is divergent.

Example 4. *Test for convergence of the series* $1+\frac{2}{3}+\frac{3}{4}+\frac{4}{5}+\ldots+\frac{n}{n+1}+\ldots\infty$

Solution. $$\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{n}{n+1} = \lim_{n\to\infty} \frac{1}{1+\frac{1}{n}} = 1 \neq 0$$

Hence, by Cauchy's Fundamental Test for divergence the series is divergent.

Example 5. *Test for convergence the series* $1+\frac{3}{5}+\frac{8}{10}+\frac{15}{17}+\ldots+\frac{2^n-1}{2^n+1}+\ldots\infty$

Solution. $\lim_{n\to\infty} u_n=\lim_{n\to\infty}\frac{2^n-1}{2^n+1}=\lim_{n\to\infty}\frac{1-\frac{1}{2^n}}{1+\frac{1}{2^n}}=1\neq 0$

Hence, by Cauchy's Fundamental Test for divergence the series is divergent.

Question. *Examine the series for convergence*

$$\frac{1}{1+2^{-1}}+\frac{2}{1+2^{-2}}+\frac{3}{1+2^{-3}}+\ldots \qquad \left(\lim_{n\to\infty} u_n\neq 0\right)$$

[Try Yourself] **Ans.** Divergent

Exercise 20.4

Examine for convergence :

1. $\frac{1}{\sqrt{2}}+\frac{2}{\sqrt{5}}+\frac{4}{\sqrt{17}}+\ldots+\frac{2^n}{\sqrt{4^n+1}}+\ldots\infty$ **Ans.** Divergent

2. $\sum_{n=1}^{\infty}\frac{n}{n+1}$ **Ans.** Divergent

3. $\sum_{n=1}^{\infty}\sqrt{\frac{n}{n+1}}$ **Ans.** Divergent

4. $\sum\cos\frac{1}{n}$ **Ans.** Divergent

5. $1+\frac{1}{2}+2+\frac{1}{3}+3+\frac{1}{4}+4+\ldots$ **Ans.** Divergent

6. $\Sigma(6-n^2)$ **Ans.** Divergent

7. $\Sigma(-2^n)$ **Ans.** Divergent

8. $\Sigma 3^{n+1}$ **Ans.** Divergent

20.14 *p*-SERIES

The series $\frac{1}{1^p}+\frac{1}{2^p}+\frac{1}{3^p}+\ldots\infty$ *is* (*i*) *convergent if* $p>1$ (*ii*) *divergent if* $p\leq 1$

Solution. Case 1 : ($p>1$)

The given series can be grouped as

$$\frac{1}{1^p}+\left(\frac{1}{2^p}+\frac{1}{3^p}\right)+\left(\frac{1}{4^p}+\frac{1}{5^p}+\frac{1}{6^p}+\frac{1}{7^p}\right)+\left(\frac{1}{8^p}+\frac{1}{9^p}+\frac{1}{10^p}+\frac{1}{11^p}+\frac{1}{12^p}+\frac{1}{13^p}+\frac{1}{14^p}+\frac{1}{15^p}\right)+\ldots$$

Now $$\frac{1}{1^p}=1 \qquad ...(1)$$

$$\frac{1}{2^p}+\frac{1}{3^p}<\frac{1}{2^p}+\frac{1}{2^p}=\frac{2}{2^p} \qquad ...(2)$$

$$\frac{1}{4^p}+\frac{1}{5^p}+\frac{1}{6^p}+\frac{1}{7^p}<\frac{1}{4^p}+\frac{1}{4^p}+\frac{1}{4^p}+\frac{1}{4^p}=\frac{4}{4^p} \qquad ...(3)$$

$$\frac{1}{8^p}+\frac{1}{8^p}+\ldots+\frac{1}{15^p}<\frac{1}{8^p}+\frac{1}{8^p}+\ldots+\frac{1}{8^p}=\frac{8}{8^p} \qquad ...(4)$$

On adding (1), (2), (3) and (4), we get :

$$\frac{1}{1^p}+\left(\frac{1}{2^p}+\frac{1}{3^p}\right)+\left(\frac{1}{4^p}+\frac{1}{5^p}+\frac{1}{6^p}+\frac{1}{7^p}\right)+\left(\frac{1}{8^p}+\frac{1}{8^p}+\ldots+\frac{1}{15^p}\right)+\ldots$$
$$<\frac{1}{1^p}+\frac{2}{2^p}+\frac{4}{4^p}+\frac{8}{8^p}+\ldots$$

$$< 1 + \left(\frac{1}{2}\right)^{p-1} + \left(\frac{1}{2}\right)^{2p-2} + \left(\frac{1}{2}\right)^{3p-3} + \ldots$$

$$< \frac{1}{1 - \left(\frac{1}{2}\right)^{p-1}} \qquad \left[\text{G.P., } r = \left(\frac{1}{2}\right)^{p-1}, \; S = \frac{1}{1-r}\right]$$

$$< \text{Finite number if } p > 1$$

Hence, the given series is convergent when $p > 1$.

Case 2 : $p = 1$

When $p = 1$, the given series becomes

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \left(\frac{1}{9} + \frac{1}{10} + \ldots + \frac{1}{16}\right) + \ldots$$

$$1 + \frac{1}{2} = 1 + \frac{1}{2} \qquad \ldots(1)$$

$$\frac{1}{3} + \frac{1}{4} > \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \qquad \ldots(2)$$

$$\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} > \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2} \qquad \ldots(3)$$

$$\frac{1}{9} + \frac{1}{10} + \ldots + \frac{1}{16} > \frac{1}{16} + \frac{1}{16} + \ldots + \frac{1}{16} = \frac{8}{16} = \frac{1}{2} \qquad \ldots(4)$$

On adding (1), (2), (3) and (4), we get

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \left(\frac{1}{9} + \frac{1}{10} + \ldots + \frac{1}{16}\right) + \ldots$$

$$> 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \ldots > 1 + \frac{n}{2} \qquad (n \to \infty)$$

$$> \infty$$

Hence, the given series is divergent when $p = 1$.

Case 3 : $p < 1$

$$\frac{1}{2^p} > \frac{1}{2}, \; \frac{1}{3^p} > \frac{1}{3}, \; \frac{1}{4^p} > \frac{1}{4} \text{ and so on}$$

therefore, $$\frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{4^p} + \ldots > 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots$$

$$> \text{divergent series } (p = 1)$$

$$\left[\text{As the series on R.H.S. } \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots\right) \text{ is divergent}\right]$$

Hence, the given series is divergent when $p < 1$.

20.15 COMPARISON TEST

If two positive terms Σu_n and Σv_n be such that

$\lim\limits_{n \to \infty} \dfrac{u_n}{v_n} = k$ (finite number), then both series converge or diverge together.

Proof. By definition of limit there exists a positive number ε, however small, such that

$$\left|\frac{u_n}{v_n} - k\right| < \varepsilon \text{ for } n > m \qquad \text{i.e., } -\varepsilon < \frac{u_n}{v_n} - k < +\varepsilon$$

$$k - \varepsilon < \frac{u_n}{v_n} < k + \varepsilon \text{ for } n > m$$

Ignoring the first m terms of both series, we have

$$k - \varepsilon < \frac{u_n}{v_n} < k + \varepsilon \quad \text{for all } n. \qquad ...(1)$$

Case 1. Σv_n is convergent, then

$$\lim_{n \to \infty} (v_1 + v_2 + \ldots + v_n) = h \text{ (say)} \qquad \text{where } h \text{ is a finite number.}$$

From (1), $u_n < (k + \varepsilon)\, v_n$ for all n.

$$\lim_{n \to \infty} (u_1 + u_2 + \ldots + u_n) < (k + \varepsilon) \lim_{n \to \infty} (v_1 + v_2 + \ldots + v_n) = (k + \varepsilon)\, h$$

Hence, Σu_n is also convergent.

Case 2. Σv_n is divergent, then

$$\lim_{n \to \infty} (v_1 + v_2 + \ldots + v_n) \to \infty \qquad ...(2)$$

Now from (1) $\qquad k - \varepsilon < \dfrac{u_n}{v_n}$

$$u_n > (k - \varepsilon)\, v_n \text{ for all } n$$

$$\lim_{n \to \infty} (u_1 + u_2 + \ldots + u_n) > (k - \varepsilon) \lim_{n \to \infty} (v_1 + v_2 + \ldots + v_n)$$

From (2) $\qquad \lim_{n \to \infty} (u_1 + u_2 + \ldots + u_n) \to \infty$

Hence, Σu_n is also divergent.

Note. For testing the convergence of a series, this Comparison Test is very useful. We choose Σv_n (p-series) in such a way that

$$\lim_{n \to \infty} \frac{u_n}{v_n} = \text{finite number.}$$

Then the nature of both the series is the same. The nature of Σv_n (p-series) is already known, so the nature of Σu_n is also known.

Example 6. *Test the series* $\sum_{n=1}^{\infty} \dfrac{1}{n + 10}$ *for convergence or divergence.*

Solution. $\qquad u_n = \dfrac{1}{n + 10}$

Let $\qquad v_n = \dfrac{1}{n}$

$$\lim_{n \to \infty} \frac{u_n}{v_n} = \lim_{n \to \infty} \frac{n}{n + 10} = \lim_{n \to \infty} \frac{1}{1 + \dfrac{10}{n}} = 1 = \text{finite number.}$$

According to Comparison Test both series converge or diverge together, but Σv_n is divergent as $p = 1$.

$\therefore \ \Sigma u_n$ is also divergent.

Example 7. *Test for convergence the series* $\frac{1}{1\cdot 2\cdot 3}+\frac{3}{2\cdot 3\cdot 4}+\frac{5}{3\cdot 4\cdot 5}+\ldots\infty$

Solution. $u_n=\frac{2n-1}{n\,(n+1)\,(n+2)}=\frac{1}{n^2}\frac{2-\frac{1}{n}}{\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)}$

Let $v_n=\frac{1}{n^2}$ $\qquad \lim_{n\to\infty}\frac{u_n}{v_n}=\lim_{n\to\infty}\frac{2-\frac{1}{n}}{\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)}=2=$ finite number.

According to Comparison Test both series converge or diverge together, but Σv_n is convergent as $p=2$.

$\therefore \ \Sigma u_n$ is also convergent.

Example 8. *Test the following for convergence :* $\sum_{1}^{\infty}\frac{\sqrt[3]{3n^2+1}}{\sqrt[4]{4n^3+2n+7}}$

Solution. $u_n=\frac{(3n^2+1)^{1/3}}{(4n^3+2n+7)^{1/4}}=\frac{3^{1/3}\,n^{2/3}\left[1+\frac{1}{3n^2}\right]^{1/3}}{n^{3/4}\left[4+\frac{2}{n^2}+\frac{7}{n^3}\right]^{1/4}}=\frac{3^{1/3}\left[1+\frac{1}{3n^2}\right]^{1/3}}{n^{1/12}\left[4+\frac{2}{n^2}+\frac{7}{n^3}\right]^{1/4}}$

Let $v_n=\frac{1}{n^{1/12}}$

$$\lim_{n\to\infty}\frac{u_n}{v_n}=\lim_{n\to\infty}\frac{3^{1/3}\left[1+\frac{1}{3n^2}\right]^{1/3}}{\left[4+\frac{2}{n^2}+\frac{7}{n^3}\right]^{1/4}}=\frac{3^{1/3}}{\sqrt{2}}=\text{finite number.}$$

According to Comparison Test both series converge or diverge together; but Σv_n is divergent as $p=\frac{1}{12}<1$.

$\therefore \ \Sigma u_n$ is also divergent.

Example 9. *Test for convergence the series whose nth term is* $n^{\log x}$.

Solution. The series is $1^{\log x}+2^{\log x}+3^{\log x}+4^{\log x}+\ldots\infty$

$$=\frac{1}{1^{-\log x}}+\frac{1}{2^{-\log x}}+\frac{1}{3^{-\log x}}+\frac{1}{4^{-\log x}}+\ldots\infty$$

Put $-\log x=p$

$$=\frac{1}{1^p}+\frac{1}{2^p}+\frac{1}{3^p}+\frac{1}{4^p}+\ldots\infty$$

(*a*) which is convergent if $p>1$, *i.e.*, $-\log x>1$

$\Rightarrow \quad \log_e\frac{1}{x}>\log_e e \ \Rightarrow \ \frac{1}{x}>e$ or $x<\frac{1}{e}$

(b) which is divergent if $p \le 1 \Rightarrow -\log x \le 1$

$$\Rightarrow \quad \log_e \frac{1}{x} \le \log_e e \Rightarrow \frac{1}{x} \le e \Rightarrow \frac{1}{e} \le x.$$

Example 10. *Test the following series for convergence* $\frac{2}{1^p} + \frac{3}{2^p} + \frac{4}{3^p} + \frac{5}{4^p} + \ldots$

Solution. Given series is $\frac{2}{1^p} + \frac{3}{2^p} + \frac{4}{3^p} + \frac{5}{4^p} + \ldots$

Here $$u_n = \frac{n+1}{n^p} = \frac{1+\frac{1}{n}}{n^{p-1}}$$

Let $$v_n = \frac{1}{n^{p-1}} \quad \therefore \quad \frac{u_n}{v_n} = \frac{1+\frac{1}{n}}{n^{p-1}} \times \frac{n^{p-1}}{1} = 1 + \frac{1}{n}$$

$$\lim_{n \to \infty} \frac{u_n}{v_n} = 1$$

Therefore, both the series are either convergent or divergent.

But Σv_n is convergent if $p - 1 > 1$, *i.e.*, if $p > 2$

and is divergent if $p - 1 \le 1$, *i.e.*, if $p \le 2$

$\therefore$ The given series is convergent if $p > 2$ and divergent if $p \le 2$.

Example 11. *Test the series for convergence* $1 + \frac{1}{2^2} + \frac{2^2}{3^3} + \frac{3^3}{4^4} + \ldots$

Solution. Let us consider $\frac{1}{2^2} + \frac{2^2}{3^3} + \frac{3^3}{4^4} + \ldots..$

Here $u_n = \frac{n^n}{(n+1)^{n+1}} = \frac{n^n}{(n+1)\,(n+1)^n} = \frac{n^n}{n\left(1+\frac{1}{n}\right) \times n^n \left(1+\frac{1}{n}\right)^n} = \frac{1}{n\left(1+\frac{1}{n}\right)\left(1+\frac{1}{n}\right)^n}$

$$\frac{u_n}{v_n} = \frac{1}{\left(1+\frac{1}{n}\right)\left(1+\frac{1}{n}\right)^n}. \quad \text{Let } v_n = \frac{1}{n}$$

$$\lim_{n \to \infty} \frac{u_n}{v_n} = \lim_{n \to \infty} \frac{1}{\left(1+\frac{1}{n}\right)\left(1+\frac{1}{n}\right)^n} = \frac{1}{e} = \text{finite quantity.}$$

Hence, either both the series are convergent or both the series are divergent. But Σv_n is divergent as $p = 1$.

$\therefore$ The given series is divergent.

Exercise 20.5

Examine the convergence or divergence of the following series :

1. $2 + \frac{3}{2} \cdot \frac{1}{4} + \frac{4}{3} \cdot \frac{1}{4^2} + \frac{5}{4} \cdot \frac{1}{4^3} + \ldots \infty$ **Ans.** Convergent

2. $1 + \frac{1 \cdot 2}{1 \cdot 3} + \frac{1 \cdot 2 \cdot 3}{1 \cdot 3 \cdot 5} + \frac{1 \cdot 2 \cdot 3 \cdot 4}{1 \cdot 3 \cdot 5 \cdot 7} + \ldots \infty$ **Ans.** Convergent

3. $1 + \frac{3}{\lfloor 2} + \frac{5}{\lfloor 3} + \frac{7}{\lfloor 4} + \ldots \infty$ **Ans.** Convergent

4. $\frac{1}{1\cdot 2}+\frac{2}{3\cdot 4}+\frac{3}{5\cdot 6}+\ldots\infty$ **Ans.** Divergent

5. $\frac{1}{1\cdot 3}+\frac{2}{3\cdot 5}+\frac{3}{5\cdot 7}+\ldots\infty$ **Ans.** Divergent

6. $1+\frac{2^2}{\lfloor 2}+\frac{3^2}{\lfloor 3}+\frac{4^2}{\lfloor 4}+\ldots\infty$ **Ans.** Convergent

7. $\frac{1}{1+2}+\frac{2}{1+2^2}+\frac{3}{1+2^3}+\ldots$ **Ans.** Convergent

8. $1+\frac{1}{2^2}+\frac{2^2}{3^3}+\frac{3^3}{4^4}+\ldots\infty$ **Ans.** Divergent

9. $\frac{1}{3}+\frac{\lfloor 2}{3^2}+\frac{\lfloor 3}{3^3}+\ldots\infty$ **Ans.** Convergent

10. $\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}+\sqrt{n+1}}$ **Ans.** Divergent

11. $\sum_{n=1}^{\infty}\frac{2n^3+5}{4n^5+1}$ **Ans.** Convergent

12. $\sum_{n=1}^{\infty}\frac{a^n}{x^n+n^a}$ **Ans.** If $x>a$, convergent; if $x\le a$, divergent

13. $\sum_{n=1}^{\infty}\frac{\sqrt{n}}{n^2+1}$ **Ans.** Convergent

14. $\sum_{n=1}^{\infty}\sqrt{(n^2+1)}-n$ **Ans.** Divergent

15. $\sum_{n=1}^{\infty}[\sqrt{(n^4+1)}-\sqrt{(n^4-1)}]$ **Ans.** Convergent

16. $\sum_{n=1}^{\infty}\frac{\lfloor n}{5^n}$ **Ans.** Divergent

17. $\sum_{n=1}^{\infty}\frac{2^n+1}{3^n+n}$ **Ans.** Convergent

18. $\sum_{n=1}^{\infty}\frac{n^n}{\lfloor n}$ **Ans.** Convergent

19. $\sum_{n=1}^{\infty}\frac{n^2}{e^n}$ **Ans.** Convergent

20. $\sum_{n=1}^{\infty}\frac{x^n}{1+x^{2n}}$ $(x>0)$ **Ans.** If $x>1$, convergent; if $x\le 1$, divergent.

21. $\sum_{n=1}^{\infty}\frac{1}{x^n+x^{-n}}$ **Ans.** If $x\le 1$, convergent; divergent if $x=1$

22. $\sum_{n=1}^{\infty}\sin\frac{1}{n}$ **Ans.** Divergent

23. $\frac{(n^2+n-1)^{1/2}}{(n^5-2)^{1/2}}$ **Ans.** Convergent

20.16 D'ALEMBERT'S RATIO TEST

Statement. *If* Σu_n *is a positive term series such that* $\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=k$ *then*

(*i*) *the series is convergent if* $k<1$ (*ii*) *the series is divergent if* $k>1$

Solution.

Case 1. When $\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=k<1$

By definition of a limit, we can find a number $r\,(<1)$ such that

$$\frac{u_{n+1}}{u_n} < r \text{ for all } n \geq m \qquad \left[\frac{u_2}{u_1} < r,\ \frac{u_3}{u_2} < r,\ \frac{u_4}{u_3} < r \ldots\right]$$

Omitting the first m terms, let the series be

$$u_1 + u_2 + u_3 + u_4 + \ldots \infty$$

$$= u_1\left(1 + \frac{u_2}{u_1} + \frac{u_3}{u_1} + \frac{u_4}{u_1} + \ldots\right) = u_1\left(1 + \frac{u_2}{u_1} + \frac{u_3}{u_2}\cdot\frac{u_2}{u_1} + \frac{u_4}{u_3}\cdot\frac{u_3}{u_2}\cdot\frac{u_2}{u_1} + \ldots \infty\right)$$

$$< u_1\,(1 + r + r^2 + r^3 + \ldots \infty) \qquad (r < 1)$$

$$= \frac{u_1}{1 - r}, \text{ which is a finite quantity.}$$

Hence, Σu_n is convergent.

Case 2. When $\lim\limits_{n \to \infty} \frac{u_{n+1}}{u_n} = k > 1$

By definition of limit, we can find a number m such that $\frac{u_{n+1}}{u_n} \geq 1$ for all $n \geq m$

$$\frac{u_2}{u_1} > 1,\ \frac{u_3}{u_2} > 1,\ \frac{u_4}{u_3} > 1$$

Ignoring the first m terms, let the series be

$$u_1 + u_2 + u_3 + u_4 + \ldots \infty$$

$$= u_1\left(1 + \frac{u_2}{u_1} + \frac{u_3}{u_1} + \frac{u_4}{u_1} + \ldots\right) = u_1\left(1 + \frac{u_2}{u_1} + \frac{u_3}{u_2}\cdot\frac{u_2}{u_1} + \frac{u_4}{u_3}\cdot\frac{u_3}{u_2}\cdot\frac{u_2}{u_1} + \ldots \infty\right)$$

$$\geq u_1\,(1 + 1 + 1 + 1 \ldots \text{ to } n \text{ terms}) = nu_1$$

$$\lim_{n \to \infty}\ (u_1 + u_2 + \ldots + u_n) = u_r$$

$$\lim_{n \to \infty}\ S_n \geq \lim_{n \to \infty}\ nu_1 = \infty$$

Hence, Σu_n is divergent.

Note. When $\frac{u_{n+1}}{u_n} = 1 \quad (k = 1)$,

The ratio test fails.

For Example. Consider the series whose nth term $= \frac{1}{n}$

$$\lim_{n \to \infty} \frac{u_{n+1}}{u_n} = \lim_{n \to \infty} \frac{\frac{1}{n+1}}{\frac{1}{n}} = \lim_{n \to \infty} \frac{n}{n+1} = \lim_{n \to \infty} \frac{1}{1 + \frac{1}{n}} = 1 \qquad \ldots(1)$$

Consider the second series whose nth term is $\frac{1}{n^2}$.

$$\lim_{n \to \infty} \frac{u_{n+1}}{u_n} = \lim_{n \to \infty} \frac{\frac{1}{(n+1)^2}}{\frac{1}{n^2}} = \lim_{n \to \infty} \left(\frac{n}{n+1}\right)^2 = 1 \qquad \ldots(2)$$

Thus, from (1) and (2) in both cases $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = 1$

But we know that the first series is divergent as $p = 1$.

The second series is convergent as $p = 2$.

Hence, when $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = 1$, the series may be convergent or divergent.

Thus, ratio test fails when $k = 1$.

Example 12. *Test for convergence the series whose nth term is* $\frac{n^2}{2^n}$.

Solution. $u_n = \frac{n^2}{2^n}$, $u_{n+1} = \frac{(n+1)^2}{2^{n+1}}$

$$\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{(n+1)^2}{2^{n+1}} \cdot \frac{2^n}{n^2} = \lim_{n\to\infty} \frac{1}{2}\left(1+\frac{1}{n}\right)^2 = \frac{1}{2} < 1$$

Hence, the series is convergent by D'Alembert's Ratio Test.

Example 13. *Test for convergence the series whose nth term is* $\frac{2^n}{n^3}$.

Solution. $u_n = \frac{2^n}{n^3}$, $u_{n+1} = \frac{2^{n+1}}{(n+1)^3}$

By D'Alembert's Ratio Test

$$\frac{u_{n+1}}{u_n} = \frac{2^{n+1}}{(n+1)^3} \cdot \frac{n^3}{2^n} = \frac{2}{\left(1+\frac{1}{n}\right)^3} \qquad \lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{2}{\left(1+\frac{1}{n}\right)^3} = 2 > 1$$

Hence, the series is divergent.

Exercise 20.6

Test for convergence the series :

1. $\sum_{n=1}^{n} \frac{n^2}{3^n}$ **Ans.** Convergent

2. $\sum_{n=1}^{n} \frac{n\,!}{n^n}$ **Ans.** Convergent

3. $\left(\frac{1}{3}\right)^2 + \left(\frac{1\cdot 2}{3\cdot 5}\right)^2 + \left(\frac{1\cdot 2\cdot 3}{3\cdot 5\cdot 7}\right)^2 + \ldots \infty$ **Ans.** Convergent

4. $\frac{2}{1} + \frac{2\cdot 5\cdot 8}{1\cdot 5\cdot 9} + \frac{2\cdot 5\cdot 8\cdot 11}{1\cdot 5\cdot 9\cdot 13} + \ldots \infty$ **Ans.** Convergent

5. $\sum_{n=1}^{n} \frac{n\,!\,.\,2^n}{n^n}$ **Ans.** Convergent

6. $\sum_{n=1}^{\infty} \frac{x^{n-1}}{n3^n}$ **Ans.** Convergent if $x > 3$ Deavergent if $x < 3$.

7. Prove that, if $u_{n+1} = \frac{k}{1+u_n}$, where $k > 0$, $u_1 > 0$, then the series Σu_n converges to the positive root of the equation $x^2 + x = k$.

8. Prove that, if $u_{n+1} + \sqrt{k+u_n}$, where $k > 0$, $u_1 > 0$, then the series Σu_n converges to the positive root of the equation $x^2 - x = k$.

9. Prove that, if $u_{n+1} = u_n^2 + k - k^2$, where $u_1 > 0$ and u_n tends to a finite limit then l must be either k or $1 - k$.

20.17 RAABE'S TEST (HIGHER RATIO TEST)

If Σu_n *is a positive term series such that* $\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}}-1\right)=k$, *then*

(*i*) *the series is convergent if* $k>1$ (*ii*) *the series is divergent if* $k<1$.

Proof. Case I. $k > 1$

Let p be such that $k > p > 1$ and compare the given series Σu_n with $\sum \frac{1}{n^p}$ which is convergent as $p > 1$.

$$\frac{u_n}{u_{n+1}} > \frac{(n+1)^p}{n^p} \quad \text{or} \quad \left(\frac{u_n}{u_{n+1}}\right) > \left(1+\frac{1}{n}\right)^p > 1+\frac{p}{n}+\frac{p(p-1)}{\lfloor 2}\frac{1}{n^2}+\ldots$$

$$n\left(\frac{u_n}{u_{n+1}}-1\right) > p+\frac{p(p-1)}{n!}\frac{1}{n^2}+\ldots$$

If $\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}}-1\right) > p$

and $k > p$ which is true as $k > p > 1$; Σu_n is convergent when $k > 1$

Case II. $k < 1$

Same steps as in Case 1.

Notes :

1. Raabe's Test fails if $k = 1$.

2. Raabe's Test is applied only when D'Alembert's Ratio Test fails.

Example 14. *Test the convergence for the series* $\frac{x}{1.2}+\frac{x^2}{3.4}+\frac{x^3}{5.6}+\frac{x^4}{7.8}+\ldots.$

Solution. $u_n=\frac{x^n}{(2n-1)\,2n}$ and $u_{n+1}=\frac{x^{n+1}}{(2n+1)\,(2n+2)}$

$$\frac{u_{n+1}}{u_n}=\frac{x^{n+1}}{(2n+1)\,(2n+2)}\times\frac{(2n-1)2n}{x^n}=\frac{x\left(1-\frac{1}{2n}\right)\cdot}{\left(1+\frac{1}{2n}\right)\left(1+\frac{2}{2n}\right)},$$

$$\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=x$$

(*i*) If $x < 1$, Σu_n is convergent (*ii*) If $x > 1$, Σu_n is divergent (*iii*) If $x = 1$, Test fails

Let us apply Rabee's Test when $x = 1$

$$\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}}-1\right)=\lim_{n\to\infty} n\left[\frac{(2n+1)(2n+2)}{2n(2n-1)}-1\right]=\lim_{n\to\infty} n\left[\frac{(2n+1)(2n+2)-2n(2n-1)}{2n(2n-1)}\right]$$

$$=\lim_{n\to\infty} n\left[\frac{(8n+2)}{(2n)(2n-1)}\right]=\lim_{n\to\infty}\frac{\left(2+\frac{2}{4n}\right)}{1\left(1-\frac{1}{2n}\right)}=2$$

So the series is convergent.

Hence we can say that the given series is convergent if $x \le 1$ and divergent. if $x > 1$. **Ans.**

Example 15. *Test the following series for convergence* $\Sigma \dfrac{1}{\sqrt{n+1}-1}$

Solution.
$$u_n = \frac{1}{\sqrt{n+1}-1}, \quad u_{n+1} = \frac{1}{\sqrt{n+2}-1}$$
$$\frac{u_n}{u_{n+1}} = \frac{\sqrt{n+2}-1}{\sqrt{n+1}-1}$$
$$\lim_{n\to\infty} \frac{u_n}{u_{n+1}} = \lim_{n\to\infty} \frac{\sqrt{1+\frac{2}{n}}-\frac{1}{n^2}}{\sqrt{1+\frac{1}{n}}-\frac{1}{n^2}} = 1$$

D'Alembert's test fails. Let us apply Raabe's test.

$$\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}}-1\right) = \lim_{n\to\infty} n\left(\frac{\sqrt{n+2}-1}{\sqrt{n+1}-1}-1\right)$$
$$= \lim_{n\to\infty} n\left[\frac{[\sqrt{n+2}-1-\sqrt{n+1}+1}{\sqrt{n+1}-1}\right] = \lim_{n\to\infty} n\left[\frac{\sqrt{n+2}-\sqrt{n+1}}{\sqrt{n+1}-1}\right]$$
$$= \lim_{n\to\infty} n\left[\frac{\sqrt{1+\frac{2}{n}}-\sqrt{1+\frac{1}{n}}}{\sqrt{1+\frac{1}{n}}-\frac{1}{n^2}}\right] = 0 < 1$$

Hence, Σu_n is divergent.

Example 16. *Test for convergence the series for positive values of x :*
$$1 + \frac{2}{5}x + \frac{6}{9}x^2 + \frac{14}{17}x^3 + \ldots + \frac{2^n-2}{2^n+1}x^{n-1} + \ldots$$

Solution. $u_n = \dfrac{2^n-2}{2^n+1}x^{n-1}$ and $u_{n+1} = \dfrac{2^{n+1}-2}{2^{n+1}+1}x^n$

By D'Alembert's Ratio Test
$$\frac{u_{n+1}}{u_n} = \left(\frac{2^{n+1}-2}{2^{n+1}+1}x^n\right)\left(\frac{2^n+1}{2^n-2}\frac{1}{x^{n-1}}\right)$$
$$\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{2^{n+1}\left(1-\frac{1}{2^n}\right)x}{2^{n+1}\left(1+\frac{1}{2^{n+1}}\right)}\left[\frac{2^n\left(1+\frac{1}{2^n}\right)}{2^n\left(1-\frac{1}{2^{n-1}}\right)}\right]$$
$$\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = x$$

(*i*) If $x < 1$, Σu_n is convergent. (*ii*) If $x > 1$, Σu_n is divergent.

(*iii*) If $x = 1$, test fails.

Let us apply Raabe's Test when $x = 1$

$$\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = \lim_{n\to\infty} n\left[\left(\frac{2^n - 2}{2^n + 1}\right)\left(\frac{2^{n+1} + 1}{2^{n+1} - 2}\right) - 1\right]$$
$$= \lim_{n\to\infty} n\,[1 - 1] = 0 < 1$$

so the series is divergent.

Hence, the given series is divergent if $x \geq 1$ and convergent if $x < 1$.

Example 17. *Test the following series for convergence*

$$1 + \frac{3}{7}x + \frac{3\cdot 6}{7\cdot 10}x^2 + \frac{3\cdot 6\cdot 9}{7\cdot 10\cdot 13}x^3 + \frac{3\cdot 6\cdot 9\cdot 12}{7\cdot 10\cdot 13\cdot 16}x^4 + \ldots$$

Solution.

$$u_n = \frac{3\cdot 6\cdot 9\cdot 12 \ldots 3(n-1)}{7\cdot 10\cdot 13\cdot 16 \ldots (3n+1)}x^{n-1}$$

$$u_{n+1} = \frac{3\cdot 6\cdot 9\cdot 12 \ldots 3(n-1)\cdot 3n}{7\cdot 10\cdot 13\cdot 16 \ldots (3n+1)(3n+4)}x^n$$

$$\frac{u_{n+1}}{u_n} = \frac{3n}{3n+4}x, \quad \lim_{n\to\infty}\frac{u_{n+1}}{u_n} = \lim_{n\to\infty}\frac{3}{3+\frac{4}{n}}x = x$$

If $x < 1$, Σu_n is convergent. If $x > 1$, Σu_n is divergent. If $x = 1$, the test fails.

$$\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = \lim_{n\to\infty} n\left(\frac{3n+4}{3n} - 1\right) = \lim_{n\to\infty} n\,\frac{4}{3n} = \frac{4}{3} > 1$$

Hence, the series is convergent if $x \leq 1$ and divergent if $x > 1$.

Example 18. *Test the following series for convergence*

$$\frac{1}{2}x + x^2 + \frac{9}{8}x^3 + x^4 + \frac{25}{32}x^4 + \ldots \infty$$

Solution.

$$u_n = \frac{n^2\cdot x^n}{2^n}, \quad u_{n+1} = \frac{(n+1)^2\cdot x^{n+1}}{2^{n+1}}$$

$$\frac{u_{n+1}}{u_n} = \frac{(n+1)^2 x^{n+1}}{2^{n+1}}\cdot\frac{2^n}{n^2 x^n} = \left(\frac{n+1}{n}\right)^2\frac{x}{2}$$

$$\lim_{n\to\infty}\frac{u_{n+1}}{u_n} = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^2\frac{x}{2} = \frac{x}{2}$$

(*i*) If $\frac{x}{2} < 1$ or $x < 2$, Σu_n is convergent. (*ii*) If $\frac{x}{2} > 1$ or $x > 2$, Σu_n is divergent.

(*iii*) If $\frac{x}{2} = 1$ or $x = 2$ the test fails.

Let us apply Raabe's test.

$$n\left(\frac{u_n}{u_{n+1}} - 1\right) = n\left[\frac{n^2}{(n+1)^2}\frac{2}{2} - 1\right] = n\left[\frac{n^2 - n^2 - 2n - 1}{(n+1)^2}\right] = \frac{-2n^2 - n}{(n+1)^2}$$

$$\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = \lim_{n\to\infty}\frac{-2-\frac{1}{n}}{\left(1+\frac{1}{n}\right)^2} = -2 < 1$$

Hence, Σu_n is divergent if $x \geq 2$, and convergent if $x < 2$.

Example 19. *Show that the series* $\frac{1}{x} + \frac{\lfloor 2}{x(x+1)} + \frac{\lfloor 3}{x(x+1)(x+2)} + \ldots$ *converges if* $x > 2$ *and diverges if* $x < 2$.

Solution.

$$u_n = \frac{\lfloor n}{x(x+1)(x+2)\ldots(x+n-1)}$$

$$u_{n+1} = \frac{\lfloor n+1}{x(x+1)(x+2)\ldots(x+n-1)(x+n)}$$

By D'Alembert's test

$$\frac{u_{n+1}}{u_n} = \frac{n+1}{(x+n)}, \quad \lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \frac{1+\frac{1}{n}}{1+\frac{x}{n}} = 1$$

Test fails. Let us apply Raabe's Test.

$$\lim_{n\to\infty}\left(\frac{u_n}{u_{n+1}} - 1\right) = \lim_{n\to\infty} n\left(\frac{x+n}{n+1} - 1\right) = \lim_{n\to\infty} n\left(\frac{x-1}{n+1}\right) = \lim_{n\to\infty} \frac{x-1}{1+\frac{1}{n}} = x-1$$

If $x - 1 > 1$ or $x > 2$, Σu_n is convergent. If $x - 1 < 1$ or $x < 2$, Σu_n is divergent.

Example 20. *Discuss the convergence of the series* $\frac{x^2}{2\log 2} + \frac{x^3}{3\log 3} + \frac{x^4}{4\log 4} + \ldots$

Solution. $\frac{x^2}{2\log 2} + \frac{x^3}{3\log 3} + \frac{x^4}{4\log 4} + \ldots$

$$u_n = \frac{x^{n+1}}{(n+1)\log(n+1)}, \quad u_{n+1} = \frac{x^{n+2}}{(n+2)\log(n+2)}$$

$$\frac{a_{n+1}}{u_n} = \frac{x^{n+1}}{(n+1)\log(n+1)} \times \frac{(n+2)\log(n+2)}{x^{n+2}}$$

$$= \frac{1}{x}\left(\frac{n+2}{n+1}\right)\frac{\log(n+2)}{\log(n+1)} = \lim_{n\to\infty} \frac{1}{x}\left(\frac{1+\frac{2}{n}}{1+\frac{1}{n}}\right)\frac{\log\left(1+\frac{2}{n}\right)}{\log\left(1+\frac{1}{n}\right)} = \frac{1}{x}$$

(*i*) when $\frac{1}{x} > 1$ *i. e.*, $x < 1$, the series is convergent

(*ii*) $x > 1$, the series is divergent. (*iii*) when $x = 1$, the test fails.

$$\frac{u_n}{u_{n+1}} = \left(\frac{n+2}{n+1}\right)\frac{\log(n+2)}{\log(n+1)} = \left(\frac{n+2}{n+1}\right)\frac{\log n + \log\left(1+\frac{2}{n}\right)}{\log n + \log\left(1+\frac{1}{n}\right)}$$

$$= \left(\frac{n+2}{n+1}\right)\frac{\log n + \frac{2}{n} - \frac{1}{2}\cdot\frac{4}{n^2} + \ldots}{\log n + \frac{1}{n} - \frac{1}{2}\cdot\frac{1}{n^2} + \ldots} = \left(\frac{n+2}{n+1}\right)\frac{1 + \frac{2}{n\log n} + \ldots}{1 + \frac{1}{n\log n} + \ldots}$$

$$= \left(\frac{n+2}{n+1}\right)\left[1+\frac{1}{n\log n}\right] = \left[1+\frac{1}{n+1}\right]\left[1+\frac{1}{n\log n}\right] = 1+\frac{1}{n\log n}$$

$$n\left[\frac{u_n}{u_{n+1}}-1\right] = n\left[1+\frac{1}{n\log n}-1\right] = \frac{1}{\log n} = 0 < 1$$

the series converges if $x \le 1$ and diverges if $x > 1$.

Example 21. *Test the series for convergence*

$$1+\frac{\alpha\cdot\beta}{1\cdot\gamma}x+\frac{\alpha(\alpha+1)\cdot\beta(\beta+1)}{1\cdot 2\cdot\gamma(\gamma+1)}x^2+\frac{\alpha(\alpha+1)(\alpha+2)\cdot\beta(\beta+1)(\beta+2)}{1\cdot 2\cdot 3\cdot\gamma(\gamma+1)(\gamma+2)}x^3+\ldots$$

Solution.

$$u_n = \frac{\alpha(\alpha+1)(\alpha+2)\ldots[\alpha+(n-1)]\cdot\beta(\beta+1)\ldots[\beta+(n-1)]}{n!\,\gamma(\gamma+1)\ldots[\gamma+(n-1)]}x^n$$

$$u_{n+1} = \frac{\alpha(\alpha+1)(\alpha+2)\ldots[\alpha+(n-1)](\alpha+n)\cdot\beta(\beta+1)\ldots[\beta+(n-1)](\beta+n)}{(n+1)!\,\gamma(\gamma+1)\ldots[\gamma+(n-1)](\gamma+n)}x^{n+1}$$

$$\frac{u_{n+1}}{u_n} = \frac{(\alpha+n)(\beta+n)}{(n+1)(\gamma+n)}x = \frac{\left(1+\frac{\alpha}{n}\right)\left(1+\frac{\beta}{n}\right)}{\left(1+\frac{1}{n}\right)\left(1+\frac{\gamma}{n}\right)}x$$

$$\lim_{n\to\infty}\frac{u_{n+1}}{u_n} = x$$

(*i*) If $x < 1$, the series is convergent.

(*ii*) If $x > 1$, the series is divergent.

(*iii*) If $x = 1$, the test fails.

Let us apply Raabe's Test.

$$n\left(\frac{u_n}{u_{n+1}}-1\right) = n\left[\frac{(n+1)(\gamma+n)}{(\alpha+n)(\beta+n)}-1\right] = n\left[\frac{n\gamma+n^2+\gamma+n-\alpha\beta-n\alpha-n\beta-n^2}{(\alpha+n)(\beta+n)}\right]$$

$$\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}}-1\right) = \lim_{n\to\infty}\frac{\gamma+\frac{\gamma}{n}+1-\frac{\alpha\beta}{n}-\alpha-\beta}{\left(\frac{\alpha}{n}+1\right)\left(\frac{\beta}{n}+1\right)} = \gamma+1-\alpha-\beta$$

(*i*) $\gamma+1-\alpha-\beta > 1$ or $\gamma > \alpha+\beta$, Σu_n is convergent.

(*ii*) $\gamma+1-\alpha-\beta < 1$ or $\gamma < \alpha+\beta$, Σu_n is divergent.

Example 22. *Test for convergence the series* $\sum\frac{4\cdot 7\cdot\ldots\cdot(3n+1)}{1\cdot 2\cdot\ldots\cdot n}x^n$

Solution. $u_{n+1} = \sum\frac{4\cdot 7\cdot\ldots(3n+1)(3n+4)}{1\cdot 2\cdot\ldots n(n+1)}x^{n+1}$

By D'Alembert's Test

$$\frac{u_{n+1}}{u_n} = \frac{(3n+4)}{n+1}x, \quad \lim_{n\to\infty}\frac{u_{n+1}}{u_n} = \lim_{n\to\infty}\frac{\left(3+\frac{4}{n}\right)x}{1+\frac{1}{n}} = 3x$$

If $3x < 1 \Rightarrow x < \frac{1}{3}$, Σu_n is convergent. If $3x > 1 \Rightarrow x > \frac{1}{3}$, Σu_n is divergent.

If $3x = 1 \Rightarrow x = \frac{1}{3}$, the test fails.

By Raabe's Test

$$\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = \lim_{n\to\infty} n\left(\frac{n+1}{n+\frac{4}{3}} - 1\right) = \lim_{n\to\infty} n\left(\frac{n+1-n-4/3}{n+4/3}\right) = \lim_{n\to\infty} \frac{-1/3}{1+\frac{4}{3n}} = -\frac{1}{3} < 1$$

Thus, Σu_n is divergent.

The series is convergent if $x < \frac{1}{3}$ and divergent if $x \geq \frac{1}{3}$.

Example 23. *Test the series for convergence* $2x + \frac{3}{8}x^2 + \frac{4}{27}x^3 + \ldots + \frac{n+1}{n^3}x^3 + \ldots \infty$

Solution. $u_n = \frac{n+1}{n^3}x^3,\ u_{n+1} = \frac{n+2}{(n+1)^3}x^{n+1}$

By D'Alembert's Test

$$\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{(n+2)\,x^{n+1}}{(n+1)^3} \cdot \frac{n^3}{(n+1)\,x^n}$$

$$= \lim_{n\to\infty} \frac{n^3(n+2)\,x}{(n+1)^4} = \lim_{n\to\infty} \frac{1\cdot\left(1+\frac{2}{n}\right)}{\left(1+\frac{1}{n}\right)^4}x = x$$

(*i*) If $x < 1$, Σu_n is convergent. (*ii*) If $x > 1$, Σu_n is divergent. (*iii*) If $x = 1$, the test fails.

Let us apply Raabe's Test when $x = 1$.

$$\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = \lim_{n\to\infty} n\left[\frac{(n+1)^4}{n^3(n+2)} - 1\right]$$

$$= \lim_{n\to\infty} n\left[\frac{n^4+4n^3+6n^2+4n+1-n^4-2n^3}{n^4+2n^3}\right]$$

$$= \lim_{n\to\infty} \frac{2n^3+6n^2+4n+1}{n^3+2n^2} = \lim_{n\to\infty} \frac{2+\frac{6}{n}+\frac{4}{n^2}+\frac{1}{n^3}}{1+\frac{2}{n}} = 2 > 1$$

Thus, the series is convergent for $x \leq 1$ and divergent if $x > 1$.

Example 24. *Test for convergence the series*

$$x + \frac{1}{2}\times\frac{x^3}{3} + \frac{1\times3}{2\times4}\times\frac{x^5}{5} + \frac{1\times3\times5}{2\times4\times6}\times\frac{x^7}{7} + \ldots$$

Solution. The given series is :

$$x + \frac{1}{2}\times\frac{x^3}{3} + \frac{1\times3}{2\times4}\times\frac{x^5}{5} + \frac{1\times3\times5}{2\times4\times6}\times\frac{x^7}{7} + \ldots \qquad \text{(G.P.)}$$

nth term of $x,\ x^3,\ x^5 \ldots$ is $x\times(x^2)^{n-1}$, *i.e.*, x^{2n-1} $[\because\ T_n = ar^{n-1}]$

$$\therefore \quad u_n = \frac{1 \times 3 \times 5 \times \ldots \times (2n-3)}{2 \times 4 \times 6 \times \ldots \times (2n-2)} \times \frac{x^{2n-1}}{2n-1}$$

$$\therefore \quad u_{n+1} = \frac{1 \times 3 \times 5 \times \ldots \times (2n-3)\,(2n-1)}{2 \times 4 \times 6 \times \ldots \times (2n-2) \times 2n} \times \frac{x^{2n+1}}{2n+1}$$

$$\therefore \quad \frac{u_{n+1}}{u_n} = \frac{(2n-1)^2}{2n\,(2n+1)} x^2 = \frac{\left(2 - \frac{1}{n}\right)^2}{2\left(2 + \frac{1}{n}\right)} x^2$$

$$\therefore \quad \lim_{n \to \infty} \left(\frac{u_{n+1}}{u_n}\right) = x^2.$$

$\therefore$ The given series is convergent if $x^2 < 1$ and divergent if $x^2 > 1$.

If $x^2 = 1$, then, D'Alembert's Test fails. Now apply Raabe's Test.

Here $$n\left(\frac{u_n}{u_{n+1}} - 1\right) = n\left[\frac{2n\,(2n+1)}{(2n-1)^2} - 1\right]$$

$$= n\left[\frac{4n^2 + 2n - 4n^2 + 4n - 1}{(2n-1)^2}\right] = \frac{n\,(6n-1)}{(2n-1)^2} = \frac{n^2\left(6 - \frac{1}{n}\right)}{n^2\left(2 - \frac{1}{n}\right)^2} = \frac{6 - \frac{1}{n}}{\left(2 - \frac{1}{n}\right)^2}$$

$$\therefore \quad \lim_{n \to \infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = \frac{6}{4} = \frac{3}{2}, \ \textit{i.e.}, > 1$$

$\therefore$ Σu_n is convergent.

Hence, Σu_n is convergent if $x^2 \le 1$ and divergent if $x^2 > 1$.

Exercise 20.7

Determine the nature of the following series :

1. $1 + \frac{\lfloor 2}{2^2} + \frac{\lfloor 3}{3^2} + \frac{\lfloor 4}{4^2} + \ldots \infty$ **Ans.** Divergent

2. $\frac{1}{1} + \frac{1 \cdot 3}{1 \cdot 4} + \frac{1 \cdot 3 \cdot 5}{1 \cdot 4 \cdot 7} + \frac{1 \cdot 3 \cdot 5 \cdot 7}{1 \cdot 4 \cdot 7 \cdot 10} + \ldots \infty$ **Ans.** Convergent

3. $1 + \frac{1+\alpha}{1+\beta} + \frac{(1+\alpha)\,(2+\alpha)}{(1+\beta)\,(2+\beta)} + \ldots \infty$ **Ans.** If $\beta - \alpha > 1$, convergent. If $\beta - \alpha \le 1$, divergent.

4. $\sum_{n=1}^{\infty} \frac{n^3}{e^n}$ **Ans.** Convergent

5. $x + \frac{2x^2}{2\,!} + \frac{3x^3}{2\,!} + \frac{4x^4}{4\,!} + \ldots \infty$ **Ans.** Convergent

6. $1 + \frac{1}{2}x + \frac{1}{5}x^2 + \frac{1}{10}x^3 + \ldots$ **Ans.** Convergent if $-1 \le x < 1$ and divergent if $|x| > 1$.

7. $1 + \frac{x}{2} + \frac{1 \cdot 3}{2 \cdot 4}x^2 + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}x^3 + \ldots \infty$ **Ans.** If $x < 1$, convergent; divergent if $x \ge 1$.

8. $x + \frac{(2x)^2}{2\,!} + \frac{(3x)^3}{3\,!} + \frac{(4x)^4}{4\,!} + \frac{(5x)^5}{5\,!} + \ldots \infty$ **Ans.** If $x < \frac{1}{e}$, convergent; divergent if $x \ge \frac{1}{e}$.

9. $1+\frac{(1!)^2}{2!}x^2+\frac{(2!)^2}{4!}x^4+\frac{(3!)^2}{6!}x^6+\ldots\infty \quad (x>0)$ **Ans.** If $x^2<4$, convergent; divergent if $x^2\geq 4$.

Find the values of x for which the following series converges :

10. $x^2(\log 2)^q+x^3(\log 3)^q+x^4(\log 4)^q+\ldots$ **Ans.** If $x<1$, convergent; divergent if $x\geq 1$.

11. $\sum_{n=1}^{\infty}\frac{(-1)^n(x+1)^n}{2^n\times n^2}$ **Ans.** If $-3<x$, convergent; if $x\leq -3$, divergent

12. $\sum_{n=0}^{\infty}\frac{(-1)^n n!\,x^n}{10^n}$

13. $\sum\frac{x^n}{2n(2n+1)}$ **Ans.** $x\leq 1$, convergent; $x>1$, divergent.

14. $\sum\frac{1\cdot 2\ldots n}{4\cdot 7\ldots(3n+1)}x^n$ **Ans.** If $0<x<3$, convergent and divergent if $x\geq 3$.

15. $\sum\frac{1\cdot 2\ldots n}{7\cdot 10\ldots(3n+4)}x^n$ **Ans.** If $0<x<3$, convergent and divergent if $x\geq 3$.

20.18 GAUSS'S TEST

Statement. *If* Σu_n *is a positive term series such that*

$$\frac{u_n}{u_{n+1}}=\alpha+\frac{\beta}{n}+\frac{\gamma}{n^2}$$

where $\alpha>0$

(i) *if* $\alpha>1$, *convergent* *if* $\alpha<1$, *divergent, whatever* β *may be*

(ii) *if* $\alpha=1$ *and* $\begin{cases}\beta>1, & \text{convergent}\\ \beta\leq 1, & \text{divergent}\end{cases}$

Example 25. *Test for convergence the series* $\frac{2^2\cdot 4^2}{3^2\cdot 5^2}+\frac{2^2\cdot 4^2\cdot 6^2}{3^2\cdot 5^2\cdot 7^2}+\frac{2^2\cdot 4^2\cdot 6^2\cdot 8^2}{3^2\cdot 5^2\cdot 7^2\cdot 9^2}+\ldots$

Solution. The given series is $\frac{2^2\cdot 4^2}{3^2\cdot 5^2}+\frac{2^2\cdot 4^2\cdot 6^2}{3^2\cdot 5^2\cdot 7^2}+\frac{2^2\cdot 4^2\cdot 6^2\cdot 8^2}{3^2\cdot 5^2\cdot 7^2\cdot 9^2}+\ldots$

$$+\frac{2^2\cdot 4^2\cdot 6^2\cdot 8^2\ldots(2n+2)^2}{3^2\cdot 5^2\cdot 7^2\cdot 9^2\ldots(2n+3)^2}+\ldots\infty$$

$$u_{n+1}=\frac{2^2\cdot 4^2\cdot 6^2\cdot 8^2\ldots(2n+2)^2(2n+4)^2}{3^2\cdot 5^2\cdot 7^2\cdot 9^2\ldots(2n+3)^2(2n+5)^2}$$

$$\frac{u_{n+1}}{u_n}=\frac{(2n+4)^2}{(2n+5)^2}=\frac{4n^2+16n+16}{4n^2+20n+25}=\frac{4+\frac{16}{n}+\frac{16}{n^2}}{4+\frac{20}{n}+\frac{25}{n^2}}$$

$$\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=\lim_{n\to\infty}\frac{4+\frac{16}{n}+\frac{16}{n^2}}{4+\frac{20}{n}+\frac{25}{n}}=1$$

D'Alembert's Test fails. Let us apply Raabe's Test.

$$\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}}-1\right)=\lim_{n\to\infty} n\left(\frac{4n^2+20n+25}{4n^2+16n+16}-1\right)$$

$$=\lim_{n\to\infty}\left(\frac{4n^2+9n}{4n^2+16n+16}\right)=\lim_{n\to\infty}\left[\frac{4+\frac{9}{n}}{4+\frac{16}{n}+\frac{16}{n^2}}\right]=1,\ \text{Raabe's Test fails}$$

Let us apply Gauss's Test

$$\frac{u_n}{u_{n+1}}=\frac{(2n+5)^2}{(2n+4)^2}=\frac{\left(1+\frac{5}{2n}\right)^2}{\left(1+\frac{2}{n}\right)^2}=\left(1+\frac{5}{n}+\frac{25}{4n^2}\right)\left(1+\frac{2}{n}\right)^{-2}$$

$$=\left(1+\frac{5}{n}+\frac{25}{4n^2}\right)\left(1-\frac{4}{n}+\frac{-2\times-3}{\lfloor 2}\frac{4}{n^2}\ldots\right)=\left(1+\frac{5}{n}+\frac{25}{4n^2}\right)\left(1-\frac{4}{n}+\frac{12}{n^2}+\ldots\right)$$

$$=1-\frac{4}{n}+\frac{12}{n^2}+\frac{5}{n}-\frac{20}{n^2}+\frac{25}{4n^2}+\ldots=1+\frac{1}{n}-\frac{7}{n^2}\qquad\left(\frac{u_n}{u_{n+1}}=\alpha+\frac{\beta}{n}+\frac{\gamma}{n^2}\right)$$

Hence, $\alpha = 1$, $\beta = 1$. Thus, the series is divergent.

20.19 CAUCHY'S INTEGRAL TEST

Statement. *A positive term series*

$$f(1)+f(2)+f(3)+\ldots+f(n)+\ldots$$

where $f(n)$ decreases as n increases, converges or diverges according to the integral

$$\int_1^\infty f(x)\,dx$$

is finite or infinite.

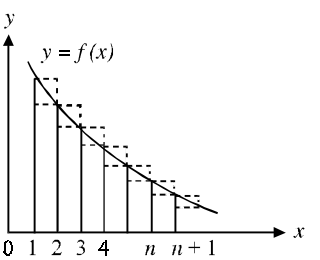
Proof. In the figure, the area under the curve from $x = 1$ to $x = n + 1$ lies between the sum of the areas of small rectangles and sum of the areas of large rectangles.

$$\Rightarrow\ f(1)+f(2)+\ldots+f(n)\geq\int_1^{n+1} f(x)\,dx\geq\ f(2)+f(3)+\ldots+f(n+1)$$

$$S_n\geq\int_1^{n+1} f(x)\,dx\geq S_{n+1}-f(1)$$

As $n\to\infty$, from the second inequality that if the integral has a finite value then $\lim_{n\to\infty} S_{n+1}$ is also finite, so $\Sigma f(n)$ is convergent.

Similarly, if the integral is infinite, then from the first inequality that $\lim_{n\to\infty} S_n\to\infty$, so the series is divergent.



Example 26. *Apply the integral test to determine the convergence of the p-series*

$$\frac{1}{1^p}+\frac{1}{2^p}+\frac{1}{3^p}+\ldots+\frac{1}{n^p}+\ldots\infty$$

Solution. (*i*) When $p > 1$ $\qquad f(x) = \dfrac{1}{x^p}$

$$\int_1^\infty f(x)\,dx = \lim_{m\to\infty} \int_1^m \frac{1}{x^p}\,dx = \lim_{m\to\infty} \left[\frac{x^{1-p}}{1-p}\right]_1^m = \lim_{m\to\infty} \frac{1}{1-p}\,[m^{1-p} - 1]$$

$$= \lim_{m\to\infty} \frac{1}{1-p}\left[\frac{1}{m^{p-1}} - 1\right] = \frac{1}{p-1}, \text{ which is finite.}$$

By Cauchy's Integral Test, the series is convergent for $p > 1$.

(*ii*) When $p < 1$

$$\int_1^\infty f(x)\,dx = \frac{1}{1-p}\left[\lim_{m\to\infty} (m^{1-p} - 1)\right] \to \infty$$

Thus, the series is divergent, if $p < 1$.

(*iii*) When $p = 1$

$$\int_1^\infty \frac{1}{x}\,dx = [\log x]_1^\infty \to \infty$$

Thus, the series is divergent.

Thus, $\Sigma \dfrac{1}{n^p}$ is convergent if $p > 1$ and divergent if $p \le 1$.

Example 27. *Examine the convergence of* $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n \log n}$.

Solution. Here $f(x) = \dfrac{1}{x \log x}$

$$\int_2^\infty \frac{1}{x \log x}\,dx = \lim_{m\to\infty} [\log\log x]_2^m = \lim_{m\to\infty} [\log\log m - \log\log 2] \to \infty$$

By Cauchy's Integral Test the series is divergent.

Example 28. *Examine the convergence of* $\displaystyle\sum_{n=1}^{\infty} ne^{-n^2}$

Solution. Here $\qquad f(x) = xe^{-x^2}$

Now $\displaystyle\int_1^\infty xe^{-x^2}\,dx = \lim_{m\to\infty} \left[\frac{e^{-x^2}}{-2}\right]_1^m = \lim_{m\to\infty} \left[\frac{e^{-m^2}}{-2} + \frac{e^{-1}}{2}\right] = \frac{e^{-1}}{2} = \frac{1}{2e}$, which is finite.

Hence, the given series is convergent.

Exercise 20.8

Examine the convergence :

1. $1 + \dfrac{x}{2} + \dfrac{x^2}{3^2} + \dfrac{x^2}{4^3} + \ldots \infty \quad (x > 0)$ **Ans.** Convergent

2. $\dfrac{2}{1^2}x + \dfrac{3^2}{2^3}x^2 + \dfrac{4^3}{3^4}x^3 + \ldots + \dfrac{(n+1)^n}{n^{n+1}}x^n + \ldots$ **Ans.** $x < 1$, convergent; $x \ge 1$, divergent.

3. $1+\frac{2^2}{3^2}+\frac{2^2\cdot 4^2}{3^2\cdot 5^2}+\frac{2^2\cdot 4^2\cdot 6^2}{3^2\cdot 5^2\cdot 7^2}+\ldots\infty$ **Ans.** Divergent

4. $\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}$ **Ans.** Divergent

5. $\sum_{n=1}^{\infty}\frac{1}{n^2+1}$ **Ans.** Convergent

6. $\sum_{n=1}^{\infty}\frac{1}{n^n}$ **Ans.** Convergent

7. $\sum_{n=1}^{\infty} n^2 e^{-n^3}$ **Ans.** Convergent

8. $\sum_{n=1}^{\infty}\frac{1}{n(\log n)^2}$ **Ans.** Convergent

20.20 CAUCHY'S ROOT TEST

Statement. *If* Σu_n *is a positive term series such that* $\lim_{n\to\infty}(u_n)^{1/n}=k$, then

(*i*) *if* $k<1$, *the series converges.* (*ii*) *if* $k>1$, *the series diverges.*

Proof. By definition of limit

$$\left|(u_n)^{1/n}-k\right|<\varepsilon \text{ for } n>m$$

$$k-\varepsilon<(u_n)^{1/n}<k+\varepsilon \text{ for } n>m$$

(*i*) $k<1$

$$k+\varepsilon<r<1$$

$$(u_n)^{1/n}<k$$

$\Rightarrow$ $u_n<k^n$

$$u_1+u_2+\ldots\infty<k+k^2+\ldots+k^n+\ldots\infty$$

$$<\frac{1}{1-k} \text{ (a finite quantity)}$$

$\therefore$ The series is convergent.

(*ii*) $k>1$

$$k-\varepsilon>1$$

$$(u_n)^{1/n}>k-\varepsilon>1$$

$$u_n>1$$

$$S_n=u_1+u_2+\ldots+u_n>n$$

$$\lim_{n\to\infty} S_n\to\infty$$

$\therefore$ The series is divergent.

(*iii*) $k=1$

If $\lim_{n\to\infty}(u_n)^{1/n}=1$, the test fails.

For example, $\Sigma u_n=\sum\frac{1}{n^p}$

$$\lim_{n\to\infty}(u_n)^{1/n}=\lim_{n\to\infty}\left(\frac{1}{n^p}\right)^{1/n}=\lim_{n\to\infty}\left(\frac{1}{n^{1/n}}\right)^{-p}=1 \text{ for all } p,\ k=1$$

But $\sum \frac{1}{n^p}$ is convergent for $p > 1$ and divergent for $p \le 1$.

Thus, we cannot say whether Σu_n is convergent or divergent for $k = 1$.

Example 29. *Examine the convergence of the series* $\sum \frac{1}{\left(1+\frac{1}{n}\right)^{n^2}}$

Solution. $u_n = \frac{1}{\left(1+\frac{1}{n}\right)^{n^2}}$, $(u_n)^{1/n} = \left[\frac{1}{\left(1+\frac{1}{n}\right)^{n^2}}\right]^{\frac{1}{n}} = \frac{1}{\left(1+\frac{1}{n}\right)^{n}}$

$$\lim_{n\to\infty} (u_n)^{1/n} = \lim_{n\to\infty} \frac{1}{\left(1+\frac{1}{n}\right)^{n}} = \frac{1}{e} < 1$$

Hence, the given series is convergent.

Example 30. *Discuss the convergence of the following series :*

$$\left(\frac{2^2}{1^2}-\frac{2}{1}\right)^{-1} + \left(\frac{3^3}{2^3}-\frac{3}{2}\right)^{-2} + \left(\frac{4^4}{3^4}-\frac{4}{3}\right)^{-3} + \ldots \infty$$

Solution. $u_n = \left[\frac{(n+1)^{n+1}}{n^{n+1}} - \frac{(n+1)}{n}\right]^{-n}$

$$[u_n]^{1/n} = \left[\left[\frac{(n+1)^{n+1}}{n^{n+1}} - \frac{n+1}{n}\right]^{-n}\right]^{\frac{1}{n}} = \left[\frac{(n+1)^{n+1}}{n^{n+1}} - \frac{n+1}{n}\right]^{-1}$$

$$\lim_{n\to\infty} (u_n)^{1/n} = \lim_{n\to\infty} \left[\left(1+\frac{1}{n}\right)^{n+1} - \left(1+\frac{1}{n}\right)\right]^{-1} = (e-1)^{-1} = \frac{1}{e-1} < 1$$

Hence, the given series is convergent.

Example 31. *Discuss the convergence of the series* $\frac{1}{2} + \frac{2}{3}x + \left(\frac{3}{4}\right)^2 x^2 + \left(\frac{4}{5}\right)^3 x^3 + \ldots \infty$

Solution. Ignoring the first term, we have :

$$u_n = \left(\frac{n+1}{n+2}\right)^n x^n$$

$$(u_n)^{1/n} = \left(\frac{n+1}{n+2}\right) x \qquad \text{[Cauchy Root Test]}$$

$$\lim_{n\to\infty} (u_n)^{1/n} = \lim_{n\to\infty} \left(\frac{1+\frac{1}{n}}{1+\frac{2}{n}}\right) x = x$$

Hence, the series converges if $x < 1$ and diverges if $x > 1$.

If $x = 1$

$$u_n = \left(\frac{n+1}{n+2}\right)^n = \left(\frac{1+\frac{1}{n}}{1+\frac{2}{n}}\right)^n \quad \lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{\left(1+\frac{1}{n}\right)^n}{\left[\left(1+\frac{2}{n}\right)^{\frac{n}{2}}\right]^2} = \frac{e}{e^2} = \frac{1}{e} \neq 0$$

$\therefore$ The series is divergent by Cauchy Fundamental theorem.

Exercise 20.9

Discuss the convergence of the following series :

1. $\sum_{n=1}^{\infty} \frac{1}{n^n}$ **Ans.** Convergent

2. $\sum_{n=1}^{\infty} \left(1+\frac{1}{n}\right)^{n^2}$ **Ans.** Divergent

3. $\sum_{n=1}^{\infty} \left(1+\frac{1}{\sqrt{n}}\right)^{-n^{3/2}}$ **Ans.** Convergent

4. $\sum_{n=1}^{\infty} \frac{1}{(\log n)^n}$ **Ans.** Convergent

5. Σn^{-k} **Ans.** If $k > 1$, convergent

6. $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^k}$ **Ans.** If $k > 1$, convergent; divergent if $k \leq 1$.

7. $\Sigma (n \log n)^{-1} (\log \log n)^{-k}$ **Ans.** If $k > 1$, convergent; divergent if $k \leq 1$.

8. $\sum \left(1-\frac{1}{n}\right)^{n^2}$ **Ans.** Convergent

9. $\sum \frac{x^n}{n^n}$ **Ans.** Convergent

10. $(a+b) + (a^2+b^2) + (a^3+b^3) + \ldots$ **Ans.** convergent if $a < 1,\ b < 1$; divergent if $a \geq 1,\ b \geq 1$.

20.21 LOGARITHMIC TEST

If Σu_n is a positive term series such that $\lim_{n\to\infty} \left(n \log \frac{u_n}{u_{n+1}}\right) = k$

(*i*) if $k > 1$, then the series is convergent. (*ii*) if $k < 1$, then the series is divergent.

Proof. (*i*) If $k > 1$

Compare Σu_n with $\sum \frac{1}{n^p}$, if $k > p > 1$, Σu_n converges.

$$\frac{u_n}{u_{n+1}} = \frac{(n+1)^p}{n^p} = \left(1+\frac{1}{n}\right)^p \qquad \ldots(1)$$

Taking logarithm of both sides of (1), we have :

$$\log \frac{u_n}{u_{n+1}} > p \log\left(1+\frac{1}{n}\right)$$

if $$\log \frac{u_n}{u_{n+1}} > p\left(\frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} - \frac{1}{4n^4} + \ldots\right)$$

if $$n \log \frac{u_n}{u_{n+1}} > p\left(1 - \frac{1}{2n} + \frac{1}{3n^2} - \frac{1}{4n^3} + \ldots\right)$$

$$\lim_{n\to\infty} n \log \frac{u_n}{u_{n+1}} > p$$

i.e., $k > p$ which is true as $k > p > 1$. $\left[\lim_{n\to\infty} n \log \frac{u_n}{u_{n+1}} = k\right]$

Hence, Σu_n is convergent.

When $p < 1$

Similarly when $p < 1$, Σu_n is divergent.

When $p = 1$, the test fails.

Example 32. *Test the convergence of the series* $x + \frac{2^2 \cdot x^2}{\lfloor 2} + \frac{3^3 \cdot x^3}{\lfloor 3} + \frac{4^4 \cdot x^4}{\lfloor 4} + \ldots \infty$

Solution. $x + \frac{2^2 \cdot x^2}{\lfloor 2} + \frac{3^3 \cdot x^3}{\lfloor 3} + \frac{4^4 \cdot x^4}{\lfloor 4} + \ldots + \frac{n^n \cdot x^n}{\lfloor n} + \ldots \infty$

$$u_n = \frac{n^n \cdot x^n}{\lfloor n}, \quad u_{n+1} = \frac{(n+1)^{n+1} \cdot x^{n+1}}{\lfloor n+1}$$

$$\frac{u_n}{u_{n+1}} = \frac{n^n \cdot x^n}{\lfloor n} \frac{\lfloor n+1}{(n+1)^{n+1} \cdot x^{n+1}} = \frac{n^n}{(n+1)^n} \frac{1}{x} = \frac{1}{\left(1+\frac{1}{n}\right)^n} \frac{1}{x}$$

$$\lim_{n\to\infty} \frac{u_n}{u_{n+1}} = \lim_{n\to\infty} \frac{1}{\left(1+\frac{1}{n}\right)^n} \frac{1}{x} = \frac{1}{e} \cdot \frac{1}{x}$$

If $\frac{1}{ex} > 1$ or $x < \frac{1}{e}$, the series is convergent.

If $\frac{1}{ex} < 1$ or $\frac{1}{e} < x$, the series is divergent. If $\frac{1}{ex} = 1$ or $x = \frac{1}{e}$, the test fails.

$$\log \frac{u_n}{u_{n+1}} = \log \frac{1}{\left(1+\frac{1}{n}\right)^n} \cdot e = \log e - \log\left(1+\frac{1}{n}\right)^n$$

$$= 1 - n \log\left(1+\frac{1}{n}\right) = 1 - n\left[\frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} + \ldots\right]$$

$$= 1 - 1 + \frac{1}{2n} - \frac{1}{3n^2} + \ldots = \frac{1}{2n} - \frac{1}{3n^2}$$

$$\lim_{n\to\infty} n \log \frac{u_n}{u_{n+1}} = \lim_{n\to\infty} \left[\frac{1}{2} - \frac{1}{3n}\right] = \frac{1}{2} < 1.$$

Thus, the series is divergent.

Example 33. *Test the convergence of the series* $1 + \frac{2x}{2\,!} + \frac{3^2x^2}{3\,!} + \frac{4^3x^3}{4\,!} + \ldots$

Solution. The given series is

$$1 + \frac{2x}{2\,!} + \frac{3^2x^2}{3\,!} + \frac{4^3x^3}{4\,!} + \ldots$$

Here $u_n = \frac{n^{n-1}\, x^{n-1}}{n\,!} \quad \therefore \quad u_{n+1} = \frac{(n+1)^n\, x^n}{(n+1)\,!}$

$$\therefore \quad \frac{u_{n+1}}{u_n} = \frac{(n+1)^n x^n}{(n+1)\,!} \times \frac{n\,!}{n^{n-1} x^{n-1}} = \frac{(n+1)^n}{(n+1)\, n^{n-1}} x$$

$$= \frac{(n+1)^{n-1}}{n^{n-1}} x = \left[\frac{n+1}{n}\right]^{n-1} x = \left[1+\frac{1}{n}\right]^{n-1} x \quad = \left(1+\frac{1}{n}\right)^n \times \left(1+\frac{1}{n}\right)^{-1} x$$

$$\therefore \qquad \lim_{n\to\infty} \frac{u_{n+1}}{u_n} = ex \qquad \left[\because \lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n = e\right]$$

By D'Alembert's Ratio Test, Σu_n is convergent if $|ex| < 1$, *i.e.*, $|x| < \frac{1}{e}$ and is divergent if $|ex| > 1$, *i.e.*, $|x| > \frac{1}{e}$.

If $ex = 1$, *i.e.*, $x = \frac{1}{e}$, this test fails.

If $x = \frac{1}{e}$

$$\frac{u_{n+1}}{u_n} = \left(1+\frac{1}{n}\right)^{n-1} \times \frac{1}{e}$$

$$\therefore \qquad \frac{u_n}{u_{n+1}} = \frac{e}{\left(1+\frac{1}{n}\right)^{n-1}}$$

$$\therefore \qquad \log\left[\frac{u_n}{u_{n+1}}\right] = \log\left[\frac{e}{\left(1+\frac{1}{n}\right)^{n-1}}\right]$$

$$= \log e - (n-1)\log\left(1+\frac{1}{n}\right) = \log e - (n-1)\left[\frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} - \ldots\right]$$

$$= 1 - \left[1 - \frac{1}{2n} + \frac{1}{3n^2} - \ldots - \frac{1}{n} + \frac{1}{2n^2} + \ldots\right] = 1 - 1 + \frac{1}{2n} - \frac{1}{3n^2} + \frac{1}{n} - \frac{1}{2n^2} - \ldots$$

$$= \frac{3}{2n} - \frac{5}{6n^2}$$

By Logarithmic Test

$$\therefore \qquad n \log \frac{u_n}{u_{n+1}} = \frac{3}{2} - \frac{5}{6n}$$

$$\therefore \qquad \lim_{n\to\infty} \left[n \log \frac{u_n}{u_{n+1}}\right] = \frac{3}{2}, \ \textit{i.e.}, > 1$$

$\therefore$ for $x = \frac{1}{e}$, Σu_n is convergent.

Hence, if $x \le \frac{1}{e}$, Σu_n is convergent and if $x > \frac{1}{e}$, Σu_n is divergent.

Exercise 20.10

Examine the convergence for the following series :

1. $\frac{1^2}{4^2} + \frac{5^2}{8^2} + \frac{9^2}{12^2} + \frac{13^2}{16^2} + \ldots \infty$ **Ans.** Convergent

2. $1+\frac{\lfloor 1}{2}x+\frac{\lfloor 2}{3^2}x^2+\frac{\lfloor 3}{4^3}x^3+\ldots\infty$ **Ans.** If $x<e$, convergent and divergent if $x\geq e$

3. $\frac{1^2}{2^2}+\frac{1^2\cdot 3^2}{2^2\cdot 4^2}x+\frac{1^2\cdot 3^2\cdot 5^2}{2^2\cdot 4^2\cdot 6^2}x^2+\ldots\infty$ **Ans.** Convergent If $x<1$, and divergent if $x\geq 1$

4. $\frac{a+x}{1\,!}+\frac{(a+2x)^2}{2\,!}+\frac{(a+3x)^3}{3\,!}+\ldots$ **Ans.** Convergent if $x<\frac{1}{e}$, divergent if $x\geq\frac{1}{e}$

20.22 DE MORGAN'S AND BERTRAND'S TEST

A series Σu_n of positive terms is convergent or divergent according as

$$\lim_{n\to\infty}\left[\left\{n\left(\frac{u_n}{n_{n+1}}-1\right)-1\right\}\log n\right]>1 \text{ or } <1.$$

Example 34. *Test for convergence the series* $1^p+\left(\frac{1}{2}\right)^p+\left(\frac{1\times 3}{2\times 4}\right)^p+\left(\frac{1\times 3\times 5}{2\times 4\times 6}\right)^p+\ldots$

Solution. The given series is :

$$1^p+\left(\frac{1}{2}\right)^p+\left(\frac{1\times 3}{2\times 4}\right)^p+\left(\frac{1\times 3\times 5}{2\times 4\times 6}\right)^p+\ldots$$

Here $$u_n=\left[\frac{1\times 3\times 5\times\ldots\times(2n-3)}{2\times 4\times 6\times\ldots\times(2n-2)}\right]^p$$

$\therefore$ $$u_{n+1}=\left[\frac{1\times 3\times 5\times\ldots\times(2n-3)\,(2n-1)}{2\times 4\times 6\times\ldots\times(2n-2)\,(2n)}\right]^p$$

$\therefore$ $$\frac{u_{n+1}}{u_n}=\left(\frac{2n-1}{2n}\right)^p=\left(1-\frac{1}{2n}\right)^p$$

$$\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=1$$

$\therefore$ D'Alembert's Test fails.

Now let us apply Raabe's Test.

Here $$n\left(\frac{u_n}{u_{n+1}}-1\right)=n\left[\left(1-\frac{1}{2n}\right)^{-p}-1\right]$$

$$=n\left[1+\frac{p}{2n}+\frac{p\,(p+1)}{8n^2}+\ldots-1\right]=\frac{p}{2}+\frac{p\,(p+1)}{8n}+\ldots$$

$\therefore$ $$\lim_{n\to\infty}n\left[\frac{u_n}{u_{n+1}}-1\right]=\frac{p}{2}$$

If $\frac{p}{2}>1$, *i.e.*, $p>2$, the series is convergent and divergent if $\frac{p}{2}<1$, *i.e.*, $p<2$.

This test fails if $\frac{p}{2}=1$, *i.e.*, $p=2$.

Now let us apply De Morgan's Test. When $p = 2$

$$n\left[\frac{u_n}{u_{n+1}} - 1\right] = 1 + \frac{3}{4n} + \ldots$$

Now $$\lim_{n\to\infty}\left[n\left(\frac{u_n}{u_{n+1}} - 1\right) - 1\right]\log n = \lim_{n\to\infty}\left[1 + \frac{3}{4n} + \ldots - 1\right]\log n$$

$$= \lim_{n\to\infty}\frac{3}{4}\left[\frac{\log n}{n} - \ldots\right] = 0 < 1 \qquad \left[\lim_{n\to\infty}\frac{\log n}{n} = 0\right]$$

$\therefore$ Σu_n is divergent when $p = 2$.

20.23 CAUCHY'S CONDENSATION TEST

If $\phi(n)$ is positive for all positive integral values of n and continually diminishes as n increases and if a be a positive integer greater than 1, then the two series $\Sigma\,\phi(n)$ and $\Sigma\, a^n\,\phi(a^n)$ are either both convergent or both divergent.

Example 35. *Test for convergence the series*

$$1 + \frac{1}{2\,(\log 2)^p} + \frac{1}{3\,(\log 3)^p} + \ldots + \frac{1}{n\,(\log n)^p} + \ldots$$

is convergent if $p > 1$ and divergent if $p = 1$ or < 1.

Solution. We apply Cauchy's Condensation Test.

Here $$\phi(n) = \frac{1}{n\,(\log n)^p}$$

$\therefore$ nth term of the second series $\Sigma\, a^n\,\phi(a^n)$ is :

$$a^n\left[\frac{1}{a^n\,(\log a^n)^p}\right] \quad i.e., \quad \frac{1}{(\log a^n)^p}\ i.e.,\ \frac{1}{(n\log a)^p}\ i.e.,\ \frac{1}{(\log a)^p}\times\frac{1}{n^p}$$

$\therefore$ the given series will be convergent or divergent if $\sum\left[\frac{1}{(\log a)^p}\times\frac{1}{n^p}\right]$ is convergent or divergent, *i.e.*, if $\sum\frac{1}{n^p}$ is convergent or divergent.

But we know that $\sum\frac{1}{n^p}$ is convergent when $p > 1$ and divergent when $p = 1$ or < 1.

Hence, the given series is convergent if $p > 1$ and divergent if $p = 1$ or < 1.

20.24 ALTERNATING SERIES

A series in which the terms are alternately negative is called the alternating series.

e.g., $$u_1 - u_2 + u_3 - u_4 + \ldots \infty$$

20.25 LEIBNITZ'S RULE FOR CONVERGENCE OF AN ALTERNATING SERIES

(*i*) Each term is numerically less than its preceeding term. (*ii*) $\lim_{n\to\infty} u_n = 0$

Example 36. *Test the following series for convergence* $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \ldots \infty$

Solution. The terms of the given series are alternately positive and negative.

$$|u_n| = \frac{1}{n},\ |u_{n+1}| = \frac{1}{n+1}$$

(*i*) $|u_{n+1}| < u_n$ as $\frac{1}{n+1} < \frac{1}{n}$ (*ii*) $\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{1}{n} = 0$

As both the conditions for convergence are satisfied, the given series is convergent by Leibnitz's rule.

Example 37. *Discuss the convergence of the series* $\sum_{n=1}^{\infty} (-1)^n \frac{n}{n^2+1}$.

Solution. The terms of the given series are alternately positive and negative; and each term is numerically less than its preceeding term.

$$\lim_{n\to\infty} \frac{n}{n^2+1} = \lim_{n\to\infty} \frac{1}{n+\frac{1}{n}} = 0$$

Hence, by Leibnitz's rule, the given series is convergent.

Example 38. *Test the convergence of the series* $\frac{1}{6} - \frac{2}{11} + \frac{3}{16} - \frac{4}{21} + \frac{5}{26} - \ldots \infty$

Solution. The terms of the given series are alternately positive and negative.

$$u_n = (-1)^{n-1} \frac{n}{5n+1}$$

$$|u_n| - |u_{n+1}| = \frac{n}{5n+1} - \frac{n+1}{5(n+1)+1} = \frac{5n^2+6n-5n^2-5n-n-1}{(5n+1)(5n+6)} = \frac{-1}{(5n+1)(5n+6)}$$

$\therefore \quad |u_n| > |u_{n+1}|$

$$\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{n}{5n+1} = \lim_{n\to\infty} \frac{1}{5+\frac{1}{n}} = \frac{1}{5} \neq 0$$

Hence, the series is not convergent. It is oscillatory.

Exercise 20.11

Discuss the convergence of the following series :

1. $1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \ldots$ **Ans.** Convergent

2. $1 - 2x + 3x^2 - 4x^3 + \ldots \infty \quad (x < 1)$ **Ans.** Convergent

3. $\frac{x}{1+x} - \frac{x^2}{1+x^2} + \frac{x^3}{1+x^3} - \frac{x^4}{1+x^4} + \ldots \infty \quad (0 < x < 1)$ **Ans.** Convergent

4. $\frac{1}{1^p} - \frac{1}{2^p} + \frac{1}{3^p} - \frac{1}{4^p} + \ldots$ **Ans.** If $p > 0$, convergent; oscillatory if $p < 0$.

5. $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{2n-1}$ **Ans.** Oscillatory

6. Show that the series $\frac{1}{x+1} - \frac{1}{x+2} + \frac{1}{x+3} - \frac{1}{x+4} + \ldots$ is convergent for all real values of x other than negative integers.

7. Prove that the series $x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots$ converges if $-1 < x \leq 1$.

20.26 ALTERNATING CONVERGENT SERIES

There are two types of alternating convergent series :

(1) Absolutely convergent series. (2) Conditionally convergent series.

Absolutely convergent series. If $u_1 + u_2 + u_3 + \ldots$ be such that $|u_1| + |u_2| + |u_3| + \ldots$ be convergent then $u_1 + u_2 + u_3 + \ldots \infty$ is called absolutely convergent.

Conditionally convergent series. If $|u_1| + |u_2| + |u_3| + ...$ be divergent and $u_1 + u_2 + u_3 + \ldots$ be convergent then $u_1 + u_2 + u_3 + \ldots \infty$ is called conditionally convergent.

Example 39. *Show that the series* $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \ldots$

is convergent but not absolutely convergent.

Solution. $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots$

The terms of the series are alternately positive and negative.

(*i*) $|u_{n+1}| < |u_n|$ as $\frac{1}{n+1} < \frac{1}{n}$ (*ii*) $\lim_{n \to \infty} u_n = 0$

Thus, the given series is convergent.

But $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots \infty$ is divergent. (p-series, $p = 1$)

Hence, the given series is conditionally convergent.

Example 40. *What can you say about the series* $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \ldots \infty$ *?*

Solution. $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \ldots$

$$|u_n| = \frac{1}{n^2}, \; |u_{n+1}| = \frac{1}{(n+1)^2}$$

(*i*) $|u_{n+1}| < |u_n|$ (*ii*) $\lim_{n \to \infty} u_n = \lim_{n \to \infty} \frac{1}{n^2} = 0$

Thus, the given series is convergent by Leibnitz's rule.

And $1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \ldots$ is also convergent. (p-series, $p = 2$)

Thus, the given series is absolutely convergent.

Example 41. *Discuss the series for convergence* $1 - \frac{1}{3} + \frac{1}{2} - \frac{1}{3^3} + \frac{1}{2^2} - \frac{1}{3^5} + \frac{1}{2^3} - \frac{1}{3^7} + \ldots$

Solution. The given series is rewritten as

$$1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \ldots - \left(\frac{1}{3} + \frac{1}{3^3} + \frac{1}{3^5} + \frac{1}{3^7} + \ldots \right)$$

$$\lim_{n \to \infty} S_n = \frac{1}{1 - \frac{1}{2}} - \frac{\frac{1}{3}}{1 - \frac{1}{3^2}} = 2 - \frac{3}{8} = 1\frac{5}{8}$$

The given series is convergent.

Again $1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \ldots + \frac{1}{3} + \frac{1}{3^3} + \frac{1}{3^5} + \ldots$

$$\lim_{n \to \infty} S_n = 2 + \frac{3}{8} = \frac{19}{8}$$

This series is also convergent. Hence, the given series is absolutely convergent.

Exercise 20.12

Discuss the convergence of the following series :

1. $1-\frac{1}{2}+\frac{1}{4}-\frac{1}{8}+\ldots\infty$ **Ans.** Absolutely convergent

2. $\frac{1}{2}-\frac{2}{5}+\frac{3}{10}-\frac{4}{17}+\ldots+\frac{(-1)^{n-1}n}{n^2+1}+\ldots\infty$ **Ans.** Conditionally convergent

3. $1-\frac{2}{3}+\frac{3}{3^2}-\frac{4}{3^3}+\ldots$ **Ans.** Absolutely convergent

4. $1+\frac{x}{2}+\frac{x^2}{3}+\frac{x^3}{4}+\ldots$ **Ans.** Divergent

5. $\frac{\sin x}{1^3}-\frac{\sin 2x}{2^3}+\frac{\sin 3x}{3^3}+\ldots$ **Ans.** Absolutely convergent

20.27 POWER SERIES IN *X*

$$a_0+a_1x+a_2x^2+\ldots+a_nx^n+\ldots$$

is a power series in x, here a's are independent of x.

Proof. $u_n=a_nx^n,\ u_{n+1}=a_{n+1}x^{n+1}$

$$\lim_{n\to\infty}\frac{u_{n+1}}{u_n}=\lim_{n\to\infty}\frac{a_{n+1}x^{n+1}}{a_nx^n}=\lim_{n\to\infty}\frac{a_{n+1}}{a_n}x$$

If $\frac{a_{n+1}}{a_n}=k$, then by D'Alembert's Ratio Test

If $|kx|<1 \implies |x|<\frac{1}{k}$, the series is convergent.

Thus, the power series is convergent if $-\frac{1}{k}<x<\frac{1}{k}$.

Thus, the *interval of the power series* is $-\frac{1}{k}$ to $\frac{1}{k}$ for *convergence*. Outside this interval the series is divergent.

Example 42. *Find the values of x for which the series* $x-\frac{x^2}{2^2}+\frac{x^3}{3^2}-\frac{x^4}{4^2}+\ldots\infty$ *converges.*

Solution. $u_n=(-1)^{n-1}\frac{x^n}{n^2},\ u_{n+1}=(-1)^n\frac{x^{n+1}}{(n+1)^2}$

$$\frac{u_{n+1}}{u_n}=-\frac{n^2}{(n+1)^2}x,\quad \lim_{n\to\infty}\frac{u_{n+1}}{u_n}=-\lim_{n\to\infty}\frac{1}{\left(1+\frac{1}{n}\right)^2}x=-x$$

By D'Alembert's Test the given series is convergent for $|x|<1$ and divergent if $|x|>1$.

At $x=+1$. The series becomes

$$1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \ldots$$

This is an alternately convergent series.

At $x = -1$. The series becomes

$$-1 - \frac{1}{2^2} - \frac{1}{3^2} - \frac{1}{4^2} + \ldots$$

This is also convergent series, $p = 2$.

Hence, the interval of convergence is $-1 \le x \le 1$.

20.28 EXPONENTIAL SERIES

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots + \frac{x^n}{n!} + \ldots$$

is convergent for all values of x.

Proof. $u_n = \dfrac{x^{n-1}}{(n-1)!}$, $u_{n+1} = \dfrac{x^n}{n!}$

$$\lim_{n \to \infty} \frac{u_{n+1}}{u_n} = \lim_{n \to \infty} \frac{x^n}{n!} \cdot \frac{(n-1)!}{x^{n-1}} = \lim_{n \to \infty} \frac{x}{n} = 0 < 1$$

Hence, by D'Alembert's Test the exponential series is convergent for all values of x.

20.29 LOGARITHMIC SERIES

$$\log (1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots + (-1)^{n-1} \frac{x^n}{n} + \ldots$$

is convergent for $-1 < x \le 1$.

Proof. $u_n = (-1)^{n-1} \dfrac{x^n}{n}$, $u_{n+1} = (-1)^n \dfrac{x^{n+1}}{n+1}$

$$\lim_{n \to \infty} \frac{u_{n+1}}{u_n} = \lim_{n \to \infty} \frac{(-1)^n x^{n+1}}{n+1} \cdot \frac{n}{(-1)^{n-1} x^n} = -\lim_{n \to \infty} \left(\frac{n}{n+1} \right) x = -\lim_{n \to \infty} \left(\frac{1}{1 + \frac{1}{n}} \right) x = -x$$

Thus, the series is convergent for $|x| < 1$ and divergent for $|x| > 1$.

At $x = 1$. The series becomes

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots \infty \text{ which is convergent.}$$

At $x = -1$. The series becomes

$$-1 - \frac{1}{2} - \frac{1}{3} - \frac{1}{4} + \ldots \infty \text{ which is divergent.}$$

20.30 BINOMIAL SERIES

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \frac{n(n-1)(n-2)}{3!} x^3 + \ldots$$

is convergent for $|x| < 1$.

Proof. $$u_r = \frac{n(n-1) \ldots (n-r+2)}{(r-1)!} x^{r-1}$$

$$u_{r+1} = \frac{n\,(n-1)\,\ldots\,(n-r+1)}{r\,!}\,x^r$$

$$\lim_{r\to\infty}\frac{u_{r+1}}{u_r} = \lim_{r\to\infty}\frac{n-r+1}{r}\,x = \lim_{r\to\infty}\left(\frac{n+1}{r}-1\right)x = -x \text{ for } r > n+1$$

If $|x| < 1$, the series is convergent by D'Alembert's Test.

Exercise 20.13

Test the convergence of the following series :

1. $1 + x + 2x^2 + 3x^3 + \ldots\, nx^n + \ldots \infty$ **Ans.** Convergent if $-1 < x > 1$

2. $\frac{1}{2} + \frac{x}{3} + \frac{x^2}{4} + \frac{x^3}{5} + \ldots + \frac{x^n}{n+2} + \ldots \infty$ **Ans.** Convergent for $-1 < x < 1$

3. $1 + \frac{1}{1\,!} + \frac{1}{2\,!} + \frac{1}{3\,!} + \ldots \infty$ **Ans.** Convergent

4. $1 + \frac{4}{1\,!} + \frac{4^2}{2\,!} + \frac{4^3}{3\,!} + \ldots \infty$ **Ans.** Convergent

5. $3 - \frac{3^2}{2} + \frac{3^3}{3} - \frac{3^4}{4} + \ldots \infty$ **Ans.** Convergent

6. $\sqrt{2} - \frac{2}{2} + \frac{2^{3/2}}{3} - \frac{2^2}{4} + \ldots \infty$ **Ans.** Convergent

7. $\frac{1}{1-x} + \frac{1}{2\,(1-x)^2} + \frac{1}{3\,(1-x)^3} + \ldots \infty$ **Ans.** Convergent if $x \neq 1$

20.31 UNIFORM CONVERGENCE

If for a given $\varepsilon > 0$, a number N can be found *independent of* x, such that for every x in the interval (a, b), the series is said to be uniformly convergent in the interval (a, b).

Example 43. *Discuss the uniform convergence of the series*

$$1 + x + x^2 + \ldots \infty$$

Solution. $$S_n = 1 + x + x^2 + \ldots + x^n = \frac{1-x^n}{1-x}$$

$$S_\infty(x) = \lim_{n\to\infty}\frac{1-x^n}{1-x} = \frac{1}{1-x} \text{ for } |x| < 1$$

$$|S_\infty(x) - S_n(x)| = \left|\frac{1}{1-x} - \frac{1-x^n}{1-x}\right| = \left|\frac{x^n}{1-x}\right| = \frac{|x|^n}{1-x} < \varepsilon$$

if $$|x|^n < \varepsilon\,(1-x)$$

Let $$|x|^N = \varepsilon\,(1-x) \implies N = \frac{\log \varepsilon\,(1-x)}{\log |x|}$$

In the interval $\left(-\frac{1}{2}, \frac{1}{2}\right)$, N has maximum value. $N = \dfrac{\log \frac{\varepsilon}{2}}{\log \frac{1}{2}}$

Hence, the given series is uniformly convergent in the interval $\left(-\frac{1}{2}, \frac{1}{2}\right)$.

Note. The series is convergent in $(-1, 1)$ but not uniformly convergent.

20.32. ABEL'S TEST

If $v_n(x)$ be either monotonic decreasing in n for each fixed x in (a, b) or monotonic increasing in n for each fixed x in (a, b), then $\Sigma\, a_n(x)\, v_n(x)$ is uniformly convergent in (a, b) if

(*i*) $\Sigma a_n(x)$ is uniformly convergent in (a, b).

(*ii*) There exists k such that $|v_n(x)| < k$ for all n when $a \le x < b$.

Example 44. *Prove that* $\frac{x^n}{n^3}$ *is uniformly convergent in* $(-1, 1)$.

Solution. (*i*) $\sum \frac{1}{n^3}$ is uniformly convergent. (*ii*) $|x^n| < k$ for all n when $-1 < k < 1$

Hence, $\sum \frac{x^n}{n^3}$ is uniformly convergent by Abel's Test.

Exercise 20.14

Prove that the following series are uniformly convergent in $(-1, 1)$.

1. $\sum \frac{x^n}{n^2}$ **2.** $\sum \frac{x^n}{n(n+1)}$ **3.** $\sum \frac{x^{2n}}{x^{2n}+n^2}$

4. Prove that the series

$$1 + \frac{e^{-2x}}{2^2-1} - \frac{e^{-4x}}{4^2-1} + \frac{e^{-6x}}{6^2-1} + \ldots \infty$$

is uniformly convergent with regard to x in $x \ge 0$.

20.33 BRIEF PROCEDURE FOR TESTING A SERIES FOR CONVERGENCE

There are four types of series :

(1) Alternating series (2) Positive term series

(3) Power series (4) Geometric series

(1) **Alternating Series.** Apply Leibnitz's Test.

(2) **Positive Term Series.**

(*a*) *Cauchy's Fundamental Test.* Find u_n, if $\lim_{n \to \infty} u_n \neq 0$, the series is divergent.

(*b*) *Comparison Test.* Apply Comparison Test $\lim_{n \to \infty} \frac{u_n}{v_n}$ where Σv_n is $\frac{1}{n^p}$.

(*c*) *D'Alembert's Ratio Test.* If Comparison Test is not applicable, apply D'Alembert's Ratio Test, $\lim_{n \to \infty} \frac{u_{n+1}}{u_n}$.

(*d*) *Raabe's Test.* If Ratio Test fails, apply Raabe's Test $\lim_{n \to \infty} n\left(\frac{u_n}{u_{n+1}} - 1\right)$.

(*e*) *Cauchy's Root Test.* If Raabe's Test fails apply Cauchy's Root Test $\lim_{n \to \infty} (u_n)^{1/n}$.

(*f*) *Logarithmic Test.* If Cauchy's Root Test fails, apply $\lim_{n \to \infty} \left[n \log \frac{u_n}{u_{n+1}}\right]$.

(*g*) *De Morgan's and Bertrand's Test.* Apply De Morgan's and Bertrand's Test.

(3) **Power Series.** Apply D'Alembert's Ratio Test. If this test fails, apply all the tests of (2).

(4) **Geometric Series**

Note. Gauss's Test and Integral Test are also useful, if they can be applied easily.

20.34 LIST OF THE TESTS FOR CONVERGENCE

1. Cauchy's Fundamental Test

$$\lim_{n\to\infty} u_n \neq 0,\ \Sigma u_n \text{ is divergent.}$$

2. Comparison Test

$$\lim_{n\to\infty} \frac{u_n}{v_n} = k,\ \Sigma u_n \text{ and } \Sigma v_n \text{ are of the same nature.}$$

$$\sum \frac{1}{n^p},\ p > 1, \text{ convergent; } p \leq 1, \text{ divergent.}$$

3. D'Alembert's Test

$$\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = k, \text{ if } k < 1,\ \Sigma u_n \text{ is convergent; if } k > 1,\ \Sigma u_n \text{ is divergent.}$$

4. Raabe's Test

$$\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = k, \text{ if } k > 1,\ \Sigma u_n \text{ is convergent; if } k < 1,\ \Sigma u_n \text{ is divergent.}$$

5. Cauchy's Root Test

$$\lim_{n\to\infty} (u_n)^{1/n} = k, \text{ if } k < 1,\ \Sigma u_n \text{ is convergent; if } k > 1,\ \Sigma u_n \text{ is divergent.}$$

6. Logarithmic Test

$$\lim_{n\to\infty} n \log \frac{u_n}{u_{n+1}} = k, \text{ if } k > 1,\ \Sigma u_n \text{ is convergent; if } k < 1,\ \Sigma u_n \text{ is divergent.}$$

7. De Morgan's and Bertrand's Test

$$\lim_{n\to\infty} \left[\left\{n\left(\frac{u_n}{n_{n+1}} - 1\right) - 1\right\} \log n\right] = k.$$

If $k > 1$, convergent; $k < 1$, divergent.

8. Gauss's Test

$$\frac{u_n}{u_{n+1}} = \alpha + \frac{\beta}{n} + \frac{\gamma}{n^2}$$

(*i*) (*a*) If $\alpha > 1$, Σu_n is convergent

(*b*) If $\alpha < 1$, Σu_n is divergent.

(*ii*) If $\alpha = 1$ and $\begin{cases} \beta > 1,\ \Sigma u_n \text{ is convergent} \\ \beta < 1,\ \Sigma u_n \text{ is divergent} \end{cases}$

9. Cauchy's Condensation Test

$\Sigma \phi(n)$ and $\Sigma a^n \phi(a^n)$ are either both convergent or both divergent.

10. Leibnitz's Test

Alternately positive and negative is convergent if

(*i*) $|u_{n+1}| > |u_n|$ (*ii*) $\lim_{n\to\infty} u_n = 0$

Exercise 20.15

Test the following series for convergence :

1. $\frac{1}{1\times 2} + \frac{1}{2\times 3} + \frac{1}{3\times 4} + \ldots + \frac{1}{n(n+1)}$ **Ans.** Convergent

2. $\frac{1}{1\times2\times3}+\frac{1}{2\times3\times4}+\frac{1}{3\times4\times5}+\ldots+\frac{1}{n(n+1)(n+2)}+\ldots$ **Ans.** Convergent

3. $\sqrt{\frac{1}{2}}+\sqrt{\frac{2}{3}}+\sqrt{\frac{3}{4}}+\sqrt{\frac{4}{5}}+\ldots$ **Ans.** Divergent

4. $\sum \tan\frac{1}{n}$ **Ans.** Divergent

5. $\sum n\tan\frac{1}{n}$ **Ans.** Divergent

6. $\frac{1}{1}+\frac{1}{2\sqrt{2}}+\frac{1}{3\sqrt{3}}+\frac{1}{4\sqrt{4}}+\ldots$ **Ans.** Convergent

7. $\Sigma\,(10^{1/n}-1)$ **Ans.** Divergent

8. $\Sigma\,[\sqrt{n^3+1}-\sqrt{n^3}]$ **Ans.** Convergent

9. $\sum\frac{\sqrt[3]{2n^2-1}}{\sqrt[4]{3n^3+2n+5}}$ **Ans.** Divergent

10. $\sum\frac{1}{n^{1+\frac{1}{n}}}$ **Ans.** Divergent

11. $\sum\sin^{-1}\frac{1}{n}$ **Ans.** Divergent

12. $\sum\sin^2\frac{1}{n}$ **Hint.** $\sin^2\frac{1}{n}=\left[\frac{1}{n}-\frac{1}{3\,!\,n^3}+\ldots\right]^2$ **Ans.** Convergent

13. $\sum\frac{1}{(1)^2a+b}+\frac{2}{(2)^2a+b}+\frac{3}{(3)^2a+b}+\ldots\frac{n}{(n)^2a+b}+\ldots$ **Ans.** Divergent

14. $\sum\frac{1}{(x-d)^2}+\frac{1}{(x-2d)^2}+\frac{1}{(x-3d)^2}+\ldots$ **Ans.** Convergent

15. $\sum\frac{1}{2^{n-1}}\left(\frac{x+1}{x-2}\right)^n$ **Ans.** Convergent if x < 1 and x > 5;
Divergent if $1\le x\ge 5$

16. $\frac{2x}{x+4}+\frac{1}{2}\left(\frac{2x}{x+4}\right)^2+\frac{1}{3}\left(\frac{2x}{x+4}\right)^3+\ldots$ **Ans.** Convergent if $-\frac{4}{3}\le x<4$; divergent if $x=4$

17. $\frac{1}{1\times3\times5}+\frac{(x+2)^2}{2\times4\times6}+\frac{(x+2)^4}{3\times5\times7}+\frac{(x+2)^6}{4\times6\times8}+\ldots$ **Ans.** Convergent if $-3\le x<1$ and $x=-1$.

18. $\frac{1}{10}+\frac{\lfloor 2}{10^2}+\frac{\lfloor 3}{10^3}+\frac{\lfloor 4}{10^4}+\ldots$ **Ans.** Divergent

19. $1+\frac{\lfloor 2}{2^2}+\frac{\lfloor 3}{3^2}+\ldots$ **Ans.** Divergent

20. $\frac{ax}{2}+\frac{a^2x^2}{5}+\frac{a^3x^3}{10}+\ldots+\frac{a^nx^n}{n^2+1}+\ldots$ **Ans.** Convergent if $-1\le ax\le 1$; divergent if $|ax|>1$

21. $\sum\frac{1}{nx^n}$ **Ans.** Convergent if $1<x$; divergent if $x\ge 1$

22. $\sum(-1)^{n-1}\frac{x^{2n-1}}{(2n-1)\,!}$ **Ans.** Convergent

23. $\sum\left(\frac{n^2-1}{n^2+1}\right)x^n$ **Ans.** Convergent if $x<1$; divergent if $x\ge 1$

24. $\sum\frac{n^3+a}{2^n+a}$ **Ans.** Convergent

25. $\frac{1}{2}+\frac{x^2}{3\sqrt{2}}+\frac{x^4}{4\sqrt{3}}+\ldots$ **Ans.** Convergent if $x^2\le 1$; divergent if $x^2>1$